



GEE1: Living in the IT Era



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Living in the IT Era

Course Description

This course covers Science, Culture and Ethics of Information Technology. It likewise provides the students an understanding on the influence of information technology on modern living and human relationships as well as its uses for personal, professional, and social advancement.

The course will also provide a better understanding about different ethical and social principles in relation to Information Technology and how we can apply these principles to make sound judgement in analyzing real world issues.

Course Outcomes

The following are the expected course outcomes for GEE 1.

1. Discuss the ethical and social impact of Information Technology today.
2. Discuss the transnational and social issues relating to the use and misuse of Information Technology.
3. Develop critical thinking and logical reasoning
4. Apply the different ethical theories and principles on analyzing and solving real world issues.
5. Understand the current status of information technology in the country.

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Chapter 1 – Introduction to the Information and Communication Technology

MODULE OVERVIEW

In this chapter, we will look at the notion of Information and Communication Technology (ICT), as well as its evolution and nature. We will also learn about the practical applications of ICT in daily life, the effects of ICT in many areas of expertise, as well as current trends and innovative technologies.

MODULE LEARNING OBJECTIVES

At the end of this, students are expected to:

- Understand the evolution of technology over the years
- Explain how technology can be used in several aspects of the society
- Discuss the impact of technology in different areas of specialization
- Identify the current trends in technology

PRE-TEST 1

Answer the following questions to gauge your knowledge on the topic.

1. What is technology?
2. How do you use technology in your daily life?
3. What are some current technological trends are you aware of?

LEARNING CONTENTS (Evolutions of Information and Communication Technology)

“Information and Communication Technology” (ICT) is a term used to describe technologies in manipulating and communicating information. ICT is used especially in relation to modern telecommunication systems as they have evolved to the intensive use of computing technology, coupled with the enormous increase in communications between computers relying on the telecommunications infrastructure to do so.

The two areas, ICT and telecommunications, have increasingly merged as telecommunications moved from analog-to-digital and then to packet technologies, and as the internet has emerged to become one of the dominant data communications systems in use today.

The very first forms of ICT, together with what we have currently, make up for the technologies that we experience today. The evolution of the first ICT to our current form of ICT goes through 4 stages: Pre-mechanical, Mechanical, Electromechanical, and Electronic. These 4 stages helped ICT evolve and adapt to the latest forms of ICT.

1. Pre-mechanical

The pre-mechanical age of technology is the earliest known form of ICT that occurred between 3000 B.C. and 1450 A.D. This is when the first human communications were observed; communications included trying to use simple symbols and pictures, known as petroglyphs, which were usually carved into rock formations. These petroglyphs made up stories, messages, and warnings.

Early alphabets were also developed such as the Phoenician Alphabet. As alphabets became more popular and more commonly used for writing information down, pens and paper began to develop. It started off as marks on wet clay, but later on, paper was created out of papyrus plants.

Since people were writing a lot of information down, they needed ways to keep it all in storage. This is where the first books and libraries were developed. Egyptian scrolls were one popular way of writing down information permanently. Some people were binding information together in book form.

2. Mechanical

The mechanical age occurred between 1450 and 1840. This when we first start to see connections between our current technology and its ancestors. More technological advancements were made during this time.

Inventions included the slide rule (an analog computer used for multiplication and dividing). The Pascaline, a hugely popular mechanical computer, was created by Blaise Pascal. Charles Babbage created the difference engine, which used the method of finite differences to tabulate polynomial equations.

Other devices were developed during this period; but, unlike current calculators, these machines could only perform one sort of computation at a time. The sizes of machinery in the mechanical period and the modern day are also incomparable.

3. Electromechanical

The electromechanical age encompasses the years 1840 to 1940. We are finally coming close to certain technologies that are like what we have now. These are the preliminary stages of telecommunications.

In the early 1800s, the telegraph was created. In 1835, Morse code was created by Samuel Morse. In 1876, Alexander Graham Bell made the telephone, one of the most popular forms of communications used up until today. The first radio was developed by Guglielmo Marconi in 1894. These technologies are extremely crucial to our modern technology; this led to big advances in the information technology and telecommunications field.

After the creation of fundamental telecommunications technology, computing machines soon followed. The first large scale automatic digital computer was the Mark 1; this was created by Harvard University's staff in the 1940s. This computer was 8 ft. high x 50 ft. long x 2 ft. wide and weighed 5 tons. It was programmed using punch cards. It was from huge machines like this that people began to look at downsizing all the parts to first make them usable by

businesses and eventually in your own home.

4. Electronic (Our Current Technology)

The electronic age is what we currently live in. It can be defined as the time between 1940 and right now.

There are 4 main sections of digital computing. The first was the era of vacuum tubes and punch cards like the ENIAC and Mark 1. Rotating magnetic drums were used for internal storage.

The ENIAC (Electronic Numerical Integrator and Computer) was the first high-speed, digital computer capable of being reprogrammed to solve a full range of computing problems. This computer was designed to be used by the U.S. Army for artillery firing tables. This machine was even bigger than the Mark 1, taking up 680 square feet (about twice the area of a parking space) and weighing 30 tons. It used vacuum tubes to do its calculations.

The second generation replaced vacuum tubes with transistors, punch cards with magnetic tape, and rotating magnetic drums with magnetic cores for internal storage. During this time, high-level programming language, such as FORTRAN and COBOL, were created.

Transistors were replaced by integrated circuits in the third generation, magnetic tape was utilized in all computers, and magnetic core was replaced by metal oxide semiconductors. Around this time, a true operating system appeared, along with the advanced programming language BASIC.

The fourth and most recent iteration included CPUs (central processing units) that integrated memory, logic, and control circuits on a single chip. Personal computers were created (Apple II). In addition, a graphical user interface (GUI) was created.

LEARNING CONTENTS (Practical Application of ICT in Everyday Life)

Information and Communications Technology have multiple practical applications in everyday life. These are the following:

a) EDUCATION

Today, most schools and higher educational institutions have computers in the classroom for teachers and students. Teachers, students, researchers, and school administrators benefit from the usage of ICT.

b) BANKING

The computer is the nerve center of the banking system around the world. It functions to control the entire banking system that also includes 'Electronic Banking Services.' Electronic banking provides 24-hour services. The services include:

- a. Automated Teller Machine (ATM)
- b. Cheque Deposit

- c. Electronic Fund Transfer
- d. Direct Deposit
- e. Pay by phone system
- f. Personal computer banking/internet banking

In the banking sector, customers, businessperson, and bank administrator benefit from the usage of ICT.

c) INDUSTRY

Computers are used to facilitate production planning and control systems, to support chain management, and to help in product design in the industrial sector.

In the industrial sector, workers, researchers, and administrators benefit from the usage of ICT.

d) E-COMMERCE

E-commerce helps in boosting the economy. It makes buying-and-selling activities easier and more efficient. For this application, computers, internet, and shared software are needed.

In the e-commerce sector, customers, suppliers, and employees benefit from the usage of ICT.

	<i>BEFORE ICT (non-computerized)</i>	<i>WITH ICT (computerized)</i>
EDUCATION	<i>Traditional – Chalk and talk</i>	<i>Learning Management Systems (LMS) are used; online classes are strengthened through online systems.</i>
	<i>Using only book and paper/magazine</i>	<i>Collaboration is emphasized through digital sharing of books and other documents.</i>
BANKING SYSTEM	<i>Banking was done manually by taking deposits directly.</i>	<i>All transactions are done digitally.</i>
	<i>Transactions can only be made during working hours.</i>	<i>transaction can be done at any time and place through mobile banking, and other means.</i>
	<i>It takes time to approve any loan applications.</i>	<i>Online services, phone banking system, credit cards are available.</i>

INDUSTRY	<i>Production was slow because everything was done manually and dependent on human labor.</i>	<i>Computers and telecommunications industry became exceedingly popular and profitable since production can be increased through an all-day operation.</i>
COMMERCE	<i>Commerce is an activity of exchanging, buying, and selling commodities on a large scale, involving transportation from place to place.</i>	<i>E-commerce plays a vital role in the economic scene. This includes the electronic distribution, buying, selling, and servicing of products.</i>
	<i>Trading was done using the barter system; this later developed into currency.</i>	
	<i>Advertisement was in the form of word of mouth, billboards, and printed flyers.</i>	
	<i>Trading globally was extremely slow and expensive. Traders had to find ways to market local products in the global market.</i>	

LEARNING CONTENTS (Impact of ICT in Different Areas of Specialization)

FASTER COMMUNICATION SPEED

In the past, it took a long time for news or messages to be sent. Now with the internet, news or messages are sent via e-mail to friends, business partners, or to anyone efficiently. Social media has also increased the spread of news and information. With the increased bandwidth capability and connection speed on the internet, any information can travel fast and instantly.

LOWER COMMUNICATION COST

Using the internet is less expensive than other modes of communication such as telephone, mailing, or courier services. It enables people to gain access to enormous amounts of data at an extremely low cost. Basic internet services are also provided at little or no cost. Furthermore, the cost of connecting to the Internet is low.

RELIABLE MODE OF COMMUNICATION

Computers are a reliable communication tool. Information may be accessed and retrieved from anywhere and at any time using computers and computer networks.

EFFECTIVE SHARING OF INFORMATION

People all over the world can now share information thanks to advances in ICT. People can share and exchange opinions, news, and information on the Internet by using discussion groups, mailing lists, online forums, and social media. This facilitates knowledge sharing, which aids in the development of a knowledge-based society.

PAPERLESS ENVIRONMENT

The term "paperless environment" was coined by ICT technology. This refers to the ability of information to be stored and retrieved digitally rather than on paper. Online communications, such as e-mail, online chat, and instant messaging, also contribute to the creation of a paperless environment.

BORDERLESS COMMUNICATION

The Internet provides quick access to information, interactivity, accessibility, and versatility. It has evolved into a borderless provider of services and information. Through the internet, information and communication can cross borders.

SOCIAL PROBLEMS

ICT has various negative consequences. It has caused societal issues in society. People nowadays prefer internet communication over face-to-face interactions. Individualism and introversion grow increasingly prevalent in people as they age. Multiple studies have also shown that increased social media have a strong link with negative self-perception.

HEALTH PROBLEMS

A computer can be harmful to people if they use it for lengthy periods of time. Users of computers are also subjected to poor posture, eyestrain, and physical and emotional stress. An ergonomic computing environment might be utilized to tackle these health issues. For example, an ergonomic chair can reduce back strain and a screen filter can minimize eye strain.

LEARNING CONTENTS (Current Trends and Emerging Technologies)

Technology is expanding at such a fast pace nowadays, allowing for faster development and advancement in society.

1. Artificial Intelligence (AI) and Machine Learning

Artificial Intelligence is a method of making a computer think intelligently like the human mind. AI is carried out by studying the patterns of the human brain and by analyzing its cognitive process. Using AI is essential in the development of intelligent software and systems.

The following are tasks processed by AI-enabled devices:

- Speech Recognition
- Object Detection
- Predictive Evaluation and Anticipating Analytics
- Smart Devices

Machine Learning is a subcomponent of Artificial Intelligence. ML applications learn from previous data or previous experience through supervised or unsupervised learning. When the ML application is exposed to new data, these applications learn, grow, change, and develop by themselves. ML involves computers finding useful information without letting them know where to look. Instead, ML leverages different algorithms that learn from data using an iterative process.

2. Robotic Process Automation (RPA)

Robotic Process Automation is the application of artificial intelligence (AI) and machine learning (ML) in software capable of handling high-volume, repeatable tasks that previously required human labor. Some of these tasks include:

- Addressing queries
- Making calculations
- Maintaining records
- Making transactions

RPA is useful in manufacturing industries.

3. Edge Computing

Edge computing is designed to solve some of the latency problems caused by cloud computing. It exists “on the edge closer to where there is a need for computing. Because of this, edge computing is used in real-time or time-sensitive data processing in remote locations with limited-to-no connectivity to a central location. In these scenarios, edge computing is utilized as mini data centers.

Edge computing will increase as use of Internet of Things (IoT) devices increase.

4. Quantum Computing

Quantum computing takes advantage of quantum phenomena, including superposition and quantum entanglement. Quantum computing is applied in banking and finance, such as for issues in fraud detection and credit risk management.

This technology is also involved in preventing the spread of the coronavirus; it helps in the development of potential vaccines and the management of epidemics, thanks to its ability to easily query, monitor, analyze and act on data.

Quantum computers are now faster than regular computers; multiple companies such as Microsoft, AWS, Google, and many others are making innovations in the field of quantum computing.

5. Virtual Reality and Augmented Reality

Virtual Reality (VR), Augmented Reality (AR), and Extended Reality (ER) are the most often used technologies for environment simulation. VR immerses the user in a virtual environment, while AR augments the user's real-world circumstances.

Although this technology trend has primarily been used for gaming thus far, it has also been

used for training, as with Virtual Ship, a simulation software used to train U.S. Navy, Army, and Coast Guard ship captains.

6. Blockchain

A blockchain is a public and chronologically-ordered series of documents called blocks. Encrypted information is done through cryptography; the encrypted information ensures that user privacy is not compromised and data cannot be altered.

Unlike modern financial institutions, blockchain network information is not controlled by a centralized authority. The network's participants maintain the data, and they have the democratic ability to authorize each transaction that can occur on a blockchain network.

7. Internet of Things (IoT)

The Internet of Things (IoT) refers to the connectivity of computing devices embedded in ordinary things that allow them to send and receive data. Sensors are installed on IoT devices to collect data from their surroundings. Environmental data is used to trigger and make automated judgments in this manner.

Consumer-level application of these devices include placing them on phones and appliances, medical and fitness equipment, livestock, and more. On an industrial level, these devices can be used for machineries, shipping equipment, vehicles, robots, warehouses, and others.

But the location of these devices matters less than what they do. IoT devices “talk” to each other by sharing data and getting feedback based on said data. These data include those that are generated, analyzed, and acted on by other devices connected through the internet.

8. 5G

5G services improve the bandwidth capabilities and network latency of computer networks. This is expected to revolutionize our lives, by enabling services that rely on advanced technologies like AR, VR, and cloud-based services. Cloud-based gaming companies like Google Stadia, NVidia GeForce Now and much more will further improve their performance through 5G. 5G will also be used in factories. An example of this is the use of camera security systems using 5G; this help improve safety and traffic management, smart grid control, and smart retail.

9. Cybersecurity

Cybersecurity focuses on defending devices and services from electronic attacks by hackers. IT professionals focus on removing the offensive components of computer networks, and improving network security for network nodes, such as computers and smartphones, and database

POST-TEST 1

Exercises:

- a. Answer the following questions:
 1. What is ICT?
 2. List down the different technologies invented in different computer generations.
 3. How does ICT affect your life?

- b. Watch the videos (click the link) below and write your thoughts about the subject matter.
1. ICT in everyday life. <https://www.youtube.com/watch?v=ZDNkhRuQNks>
 2. Artificial Intelligence. <https://youtu.be/ad79nYk2keg>
 3. Robotic Process Automation. <https://youtu.be/H5VKUajGDdI>
 4. Block Chain Technology. <https://youtu.be/E2JRnFgrztM>
 5. Cyber Security. <https://youtu.be/z5nc9MDbvkW>

SUMMARY

- ICT stands for information, communication, and technology.
- The evolution of ICT goes through stages, namely: (1) Pre-mechanical, (2) Mechanical, (3) Electromechanical, and (4) Electronic.
- The different application of ICT in our everyday life embraces education, banking, industry, and e-commerce.
- The effect of ICT in different areas of specialization are the following: (1) faster communication speed, (2) lower communication cost, (3) reliable mode of communication, (4) effective sharing of information, (5) paperless environment, (6) borderless communication, among others.
- The current trends and emerging technologies include the following: (1) Artificial Intelligence, (2) Machine Learning, (3) Robotic Process Automation, (4) Edge Computing, (5) Quantum Computing, (6) Blockchain, (7) Internet of Things, (8) 5G, and (9) Cybersecurity.

REFERENCES

E-SOURCES:

- *How ict affect our daily life.* (2019, March 6). AilynVilla. <https://ailynvilla.home.blog/2019/03/04/how-ict-affect-our-daily-life/>
- Duggal, N. (2018, August 8). *Top 9 new technology trends for 2021.* Simplilearn.com. <https://www.simplilearn.com/top-technology-trends-and-jobs-article>
- Duggal, N. (2020, April 22). *What is artificial intelligence: Types, history, and future.* Simplilearn.com. <https://www.simplilearn.com/tutorials/artificial-intelligence-tutorial/what-is-artificial-intelligence>
- ICT in everyday life- Form 4. (n.d.). Information and Communication Technologies. <https://cikgushelly.blogspot.com/2012/01/ict-in-everyday-life-form-4.html>

Chapter 2 – The Web and the Internet

MODULE OVERVIEW

In this chapter, we will discuss the evolution of the Internet along with many ways to connect to the Internet, how data travels the Internet, and how the Internet works with domain names and IP Addresses. It also introduces services available on the Internet, such as email, video conferencing, telenet, file transferring, world wide web, ecommerce, newsgroup, and internet relay chat. Finally, understand the ethical issues and how you protect yourself while using the Internet.

MODULE LEARNING OBJECTIVES

At the end of this, students are expected to:

- Explore the current breakthrough technologies and disruptive innovations that have appeared over the past few years.
- Identify the different services offered through the web.
- Appreciate the evolution of world wide web.
- Understand the risk of putting your information over the internet.

PRE-TEST 2

Answer the following questions to gauge your knowledge on the topic.

1. What is the Internet? How do you use the internet in your daily life?
2. What is the precursor to the Internet?
3. How does networking affect people's data privacy?

LEARNING CONTENTS (Introduction to Internet)

The internet is a worldwide collection of networks that connects millions of businesses, government agencies, educational institutions, and individuals. Each of the networks on the Internet provides resources that add to the abundance of goods, services, and information accessible via the Internet.

The Internet has its roots in a networking project started by the Pentagon's Advanced Research Projects Agency (ARPA), an agency of the U.S> Department of Defense. ARPA's goal was to build a network that (a) allowed scientists at different physical locations to share information and work together on military and scientific projects and (b) could function even if part of the network were disabled or destroyed by a disaster such as a nuclear attack. That network, called ARPANET, became functional in September 1969, linking scientific and academic researchers across the United States.

The Internet consists of many local, regional, national, and international networks. Both public and private organization own networks on the Internet. These networks, along with phone companies, cable and satellite companies, and the government, all contribute toward the

internal structure of the Internet.

The popular term for the Internet is the “information highway”. Rather than moving through geographical space, it moves your ideas and information through cyberspace – the space of electronic movement of ideas and information.

Connecting to the Internet

Users can connect their computers and mobile devices to the Internet through wired or wireless technology and then access its services free or for a fee. With wired connections, a computer or device physically attaches via a cable or wire to a communication device such as a modem, that transmits data and other items over a transmission media to the Internet. For wireless connections, many mobile computers and devices include the necessary built-in technology so that they can transmit data and other items wirelessly.

- A wireless modem uses a wireless communication technology (such as satellite or WIFI) to connect to the Internet. Figure 1 below shows example of modem that connects to a computer and enables added functions when attached.

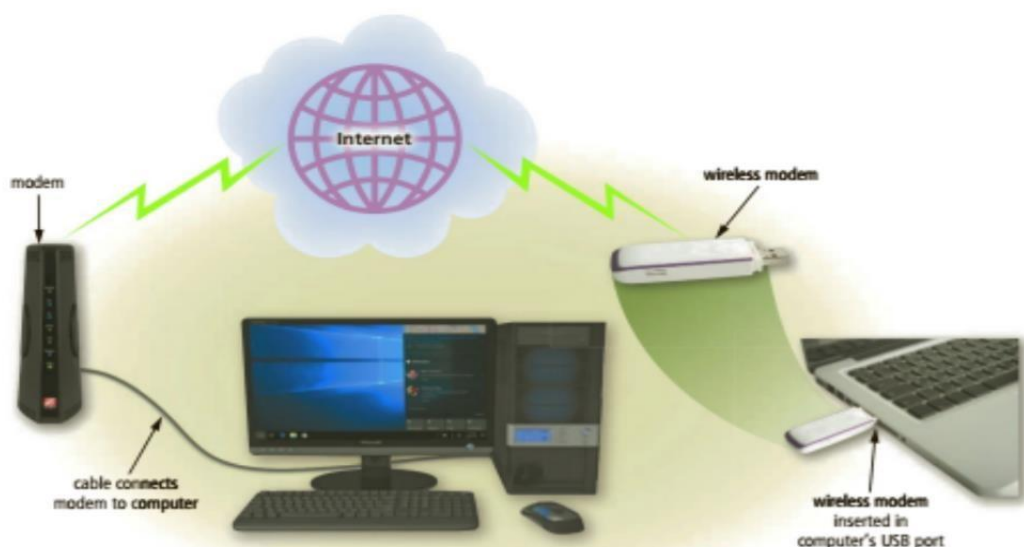


Figure 1. Using modem/s to connect computers and mobile devices to the Internet
(Source: Cengage Learning)

- Today, users often connect to the Internet via broadband Internet service because of its fast data transfer speeds and its always-on connection. Through broadband Internet service, users can download webpages quickly, play online games, communicate in real time with others, and more.
- Many public locations, such as shopping malls, coffee shops, restaurants, schools, airports, hotels, and city parks have WI-FI hot spots. **Hot Spot** is a wireless network that provides Internet connections to mobile computers and devices. Although most hot spots enable unrestricted or open access, some require that users agree to terms of service, obtain a password or perform some other action to connect the Internet.

Internet Service Providers (ISP)

An Internet Service Provider (ISP), often known as an Internet access provider, is a company

that provides free or paid Internet connection to consumers and companies. ISPs (Internet Service Providers) often charge a flat fee for an Internet connection, with consumers able to choose from a range of plans based on desired speeds, capacity, and services.

Bandwidth is a measure of a network's capacity to send and receive data. During the same period, a high bandwidth connection sends more data than a low bandwidth connection. Data sizes are commonly expressed in megabytes and gigabytes.

Megabyte (MB) is equal to approximately one million characters.

Gigabyte (GB) is equal to approximately one billion characters.

Mobile Service Providers

A mobile service provider, also known as a wireless data provider, is an Internet service provider that provides wireless Internet access to computers and mobile devices that have the necessary built-in wireless capability, wireless modems, or other communications devices that enable wireless connectivity. To communicate with a mobile service provider, an antenna attached to or integrated into a computer or device, wireless modem, or communications device commonly delivers signals over the airways.

Connected computers and gadgets use servers and clients, as well as different wired and wireless transmission mediums, to send data around the world. Your computer or device is a client on the Internet, which may access data and services on several servers. Phone lines are examples of wired transmission medium.

IP Addresses and Domain Names

IP address, which stands for Internet Protocol address, is a series of digits that uniquely identify the location of every computer or device connected to the Internet or any other network.

A domain name is a text-based name that correlates to a server's IP address, such as a web server that hosts a website. It is part of the web address that you put into the address bar of your browser to access a website.

The top-level domain (TLD) is the suffix of the domain name that specifies the kind of organization linked with the domain. The table below includes several prominent TLDs.

TLD	Intended Purpose
.biz	Businesses
.com	Commercial organization, businesses, and companies
.edu	Educational institutions
.gov	Government agencies
.mil	Military organizations
.net	Network providers or commercial companies
.org	Nonprofit organizations
.pro	Licensed professionals

- The organization that approves and controls TLDs is the **Public Technical Identifiers (PTI)**. This nonprofit organization is an affiliate of **ICANN** which stands for Internet Corporation for Assigned Names and Numbers.
- The **domain name system (DNS)** is the method that the Internet uses to store domain names and their corresponding IP addresses.

LEARNING CONTENTS (Services offered through the Internet)

Some of the services provided by the internet are as follows.

1. **Electronic Mail (E-mail).** Electronic mail is an electronic message that is transferred from one computer to another computer through the internet. Here are some of the advantages of e-mail:
 - a. It is the cheapest and fastest mail service.
 - b. you can send any form of data like text, image, video, etc.
 - c. physical presence or recipient is not needed.
 - d. It can be read anywhere in the world
2. **Telenet.** Telenet is an Internet service that allows a computer in another area to access a computer in another location. A remote computer may be accessed from another place via Telenet.
3. **File Transferring.** The Internet allows for the downloading and uploading of massive amounts of data. To upload or download files from a distant computer, we require an FTP Client application.
4. **Video Conferencing.** Video conferencing is an innovative Internet service that allows individuals in various locations to converse by observing one other.
5. **World Wide Web (WWW).** This site has millions of pages of information on a variety of topics. There are millions of information-provider websites on the internet. The Internet Address or Uniform Resource Locator identifies each website (URL).
6. **E-commerce.** It is an Internet service that allows us to sell or purchase items. We may get it using specialized websites. Electronic commerce sites are those that allow users to purchase and sell items on their websites.
7. **Electronic Fax (E-Fax).** It is a service that allows users to send and receive faxes from one computer to another. A computer with a modem, an e-fax application, and access to the internet is required to utilize the electronic fax service.
8. **Newsgroup.** A newsgroup is one of the services given by the Internet that allows users to publish notes or messages on a variety of topics.
9. **Internet Relay Chat.** It is also one of the most popular Internet services, allowing users in different regions to exchange and receive instant messages. We may use Internet Relay Chat to transmit voice or text messages.

LEARNING CONTENTS (Evolution of the World Wide Web)

Although the Internet was developed in the late 1960s, the World Wide Web (WWW) appeared in the early 1990s as an easier way to access online information using a web browser.

The World Wide Web (WWW) is a system of interconnected hypertext documents accessed through the internet. To access text, images, videos, and other multimedia tools online, one must have an internet connection. Tim Berners-Lee and Robert Cailliau proposed the use of hypertext to integrate information into the web as nodes; through a browser, users can view the hypertext. In this way, World Wide Web was designed as the first web service.

- **Web 1.0: The Read-Only Web/ The Hypertext Web**

Web 1.0 is the first generation of WWW and lasted between 1989 and 2005. Berners-Lee coined the term Web 1.0, and stated it as the read-only web. Through read-only, internet users were only reading information presented to them. It is less dynamic and less interactive in comparison to what we currently have. The primary aim of websites at this time was to make information public for anyone and set up an online presence; web was mainly focused on content delivery rather than interaction and production.

- **Web 2.0: The Read-Write Web/ The Social Web**

Web 2.0, or the read-write web, is more people-centric and participative compared to its precursor. Unlike 1.0 version, Web 2.0 allows more control to users and provides interaction. At this point, the web has introduced a business revolution in the computer industry; businesses have moved into the web platform to introduce their services to more people.

Interaction in Web 2.0 is bidirectional; businesses publish content, as do people. That is why Web 2.0 is also distinctly called as the Social Web. Blogs and other social profiles became popular at this time.

- **Web 3.0: Executable Web/ The Semantic Web**

Web 3.0 is known as the semantic web; it includes integration, automation, data, and discovery. It encourages mobility and globalization.

Through Web 3.0, users are allowed to find, share, and collaborate among each other. The semantic web allows a person or a computer to start off in one database, and then move through an unending set of databases.

From Berners-Lee, "if HTML (Hypertext Markup Language) and the Web made all the online documents look like one huge book, RDF, schema, and inference languages will make all the data in the world look like on huge database".

- **Web 4.0**

Web 4.0 is founded on the use of wireless communications. Computers and mobile devices are utilized to connect people and objects anytime and anywhere. An example of Web 4.0 technology is the GPS (Global Positioning System).

- **Web 5.0**

Currently, Web 5.0 is still progressing. This is defined as linked, emotive, and symbolic web that promotes human-computer interaction. To add, Web 5.0 is centered on emotional connections with humans. It acts similarly to a personal assistant. Technology and human contact have grown inextricably linked. In this approach, they will pave the path for a more customized experience, which will attract an increasing number of individuals.

LEARNING CONTENTS (Ethical Issue – A Glimpse of Privacy)

Computer ethics are defined as a “set of moral principles that govern an individual or a group on what is acceptable behavior while using a computer.” It governs the usage of computers and the sharing of information over the internet.

Some of the common issues of computer ethics is the violation of copyright issues, accessing personal information of other people, confidentiality and privacy issues, among other issues. These common issues are defined below.

Computer Ethics Issues

1. **Ethics and Responsible Decision-Making.** An IT student/practitioner/professional should be made aware of their individual responsibility in making ethical decisions associated with information security.
2. **Confidentiality & Privacy.** Computers can be used to frighten, deceive, or swindle victims metaphorically. Attorneys, government entities, and companies are increasingly using piles of computer-generated data to confuse their audiences legally.

The computer has made invasions of our privacy much easier and perhaps more hazardous than they were before the computer. Individuals' data is gathered and kept in a variety of digital files. The illegal commercial use of electronic data, breaches of confidentiality through revealing sensitive data to other parties, and the leaking of records to governmental authorities for investigation purposes are all possible risks to privacy.

3. **Piracy.** This is the act of illegally reproducing or disseminating copyrighted material, such as computer programs, books, music, and films, and other media.
4. **Fraud & Misuse.** The introduction of fake records into a computer system, theft of money by electronic methods, theft of financial instruments, theft of services, and theft of valuable data are all examples of computer-related fraud.
5. **Liability.** An explicit warranty includes promises and affirmations made by the software creator to the user concerning the nature and quality of the application. Express warranties can be defined by programmers or merchants. As a result, businesses must be realistic when making claims and forecasts about the capabilities, quality, and nature of their software or hardware. They should think about the legal implications of their affirmative promises, product demonstrations, and product descriptions. Every word they say might be as legally binding as if it were written down.

As a result, all agreements should be in writing to avoid responsibility. A disclaimer of explicit guarantees can protect a provider from liability for any informal, hypothetical remarks or forecasts made during the transaction.

6. **Patent and Copyright Law.** A patent can safeguard an idea's unique and confidential qualities. In comparison to copyright, obtaining a patent is extremely difficult. Complete disclosure is necessary in the case of computer software; the patent holder must publish all the specifics of a program for a skilled programmer to develop the program.

Copyright law is an extremely powerful legal instrument for safeguarding computer software, both before and after a security incident. This sort of breach might involve the theft of data, computer programs, documents, or other comparable materials. As a result, the information security professional should be conversant with the fundamentals of copyright law.

7. **Trade Secrets.** A trade secret safeguards something valuable and beneficial. This legislation safeguards the unique and private elements of ideas that are known only to the discoverer and his/her confidants. Once revealed, the trade secret is no longer a secret and can only be protected by one of the following laws. The use of trade secret legislation is especially crucial in the computer industry, where even a little head start in the creation of software or hardware can give a major competitive advantage.
8. **Sabotage.** The computer can be the object of attack in computer crimes such as the unauthorized use of computer facilities, alteration or destruction of information, data file sabotage, and vandalism against a computer system. Computers have been shot, stabbed, short-circuited, and bombed, which affects their performance,

Protecting Yourself While Using the Internet

1. **Educating Children/Teens.** For parents and or older siblings, talk to and teach your children/younger siblings about the perils of the internet and the threats of cyber predators. Tell them not to meet people they met online in person and not to chat to people they do not know. Discuss the significance of not sharing identifying information, sensitive information, images, or details about forthcoming events on the internet.

Make sure kids realize that what they see on the internet may not be accurate. Teach children that anything they put online is public unless they properly regulate the privacy settings. Spend time on the internet with your kid, learn about their favorite online sites, know their passwords, restrict the amount of time they spend online, and consider putting the computer in a public area of the house. Examine your child's PC and emails on a regular basis. Know who your kids/siblings are chatting with online.

2. **Emails, Phishing, and Malware.** When opening emails from unfamiliar individuals or sources, especially when they are unsolicited, use caution. By clicking on links or downloading attachments, you risk infecting your computer with a virus or being a victim of fraud, malware, or a scam. Some viruses can destroy your computer, while others can steal your personal information and, eventually, your identity. When you get emails that appear to be from your bank or another financial institution, be suspicious,

especially if they require you to verify or submit personal or financial information. Be

wary of scams that employ email links to take you to a website or provide you with a phone number to call. Some email links might be deceptive. Consider entering your own link to the banks and corporations, or search for them.

In general, be wary of email scammers and websites that aim to deceive you into disclosing personal information. A legitimate-looking website may be created fast. Remember that professional customer care employees will never request personal information or passwords from you. Consider not replying to unwanted emails, never click on links in these emails, and use caution if a speedy response is asked. Consider buying or downloading an excellent antivirus package that includes spyware protection.

- 3. Parental Controls.** To limit internet access to safe websites, parents should consider using parental controls through their internet service provider and/or blocking software on family computers and smartphones. If you have any questions, please contact your internet service provider. Research your options when it comes to parental controls on different items.
- 4. Passwords.** Make secure passwords with at least 10 (ten) characters and a mix of upper-case letters, lower-case letters, symbols, and digits. Personal information should not be included. When dealing with sensitive information, consider changing your password at least every 90 days (approximately 3 months). Passwords should never be left near your computer or in plain sight. Use multiple passwords for different online activity since if one is compromised, all are vulnerable. Never give out your password. If you are offered the option of setting up a password "hint" on an account, do not use something that others may simply guess.
- 5. Personal Information.** Think twice before posting anything online or sending information in an email. Anyone can see what you publish online. Sharing personal information with someone you don't know is one of the most dangerous things you can do online. Sharing sensitive information like your address, phone number, family members' names, car information, passwords, work history, credit status, social security numbers, birth date, school names, passport information, driver's license numbers, insurance policy numbers, loan numbers, credit/ debit card numbers, PIN numbers, and bank account information is dangerous and should be avoided. Consider deleting your name from websites that disclose your personal information gathered from public records with everyone on the internet (including your phone number, address, social media avatars, and images).
- 6. Photos.** Photos taken with cellphones include GPS coordinates, which allow people to decide the position of the photograph and may be used to locate you. When uploading images to online social networking sites, keep this in mind. Remember that photographs you upload online can be copied, edited, and shared with many people without your knowledge or agreement, unless you use privacy settings to limit who can

see them.

7. **Public Computers.** Avoid typing sensitive information on public computers, such as those in a public library or an internet café. Spyware may be installed on these computers that record your every keystroke. Also, you never know who may be watching your activity. Never select the feature that automatically signs you on to email or check any box to “Remember my Password” on websites.
8. **Selling.** Beware of selling and listing items in local ads or elsewhere online. Never meet someone alone. If necessary, consider meeting in a public place, like a post office or bank rather than a parking lot. Beware of posting photos taken from smartphones for online adds. You could be sharing your home address with a criminal.
9. **Shopping.** Avoid purchasing products and services from websites that do not have a secure check-out using “HTTPS (Hypertext Transfer Protocol Secure).” Pay attention to checkout page’s address line; the checkout page asks you to enter your credit card information. If the checkout page does not have an “S” after “HTTP”, consider shopping elsewhere. Be warned that certain information transferred on HTTP pages is in plain text, which computer hackers may intercept.
10. **Social.** Be wary about meeting somebody in person who you met on the internet or over email. Not everyone is truthful about their identity, age, gender, or objectives. Do your investigation using public documents if required, and consider finding reliable recommendations. If you decide to meet someone, never travel alone, notify others of your plans, meet in a public spot, and have your mobile phone nearby.
11. **Updates.** Keep your computer’s operating system, browsers, antivirus, and other software up to date with the latest or daily security patches.
12. **Webcams.** Be careful when using webcams. These devices can be high-jacked and turned on remotely. This allows others to illegally view and listen to individuals without their knowledge. Consider turning them off or disconnecting them when not in use. Limit or do not allow your children to use webcams and talk to them about the risks.
13. **Wireless.** Beware when connecting your laptops or mobile devices to any unsecured networks. Computer hackers on the same network can intercept your internet use; in some cases, they can also access files on your computer. To add, consider password protecting your home wireless network and using a personal firewall program for added protection.

POST-TEST 2

Answer the questions below.

- c. Who owns the Internet?
- d. How to use public WI-FI hotspot safely?
- e. How to register a domain name?

Watch the videos (link)

1. History of the World Wide Web - <https://youtu.be/WlryJFlyr10>
2. Uses of the Internet - <https://youtu.be/sawN1GVqDmE>
3. Tim Berners Lee: https://youtu.be/OM6XIIcm_g0

SUMMARY

Module 2 presented the current breakthrough technologies and disruptive innovations that have appeared over the past few years and discussed the different services offered through the web. It presented the evolution of world wide web from web 1.0 to web 5.0. Finally, this module listed the different ethical issues and some tips on how to protect yourself while using the Internet.

REFERENCES

E-SOURCES:

- *Chapter-5: Internet and Web-II.* (2021, July 1). INFORMATICS PRACTICES. <https://studywidpranay.wordpress.com/chapter-10-introduction-to-internet-and-web/>
- *Ethical issues.* (n.d.). Home. <https://www.niatec.iri.isu.edu/ViewPage.aspx?id=153>
- *Internet ethics - Student - ISEA.* (n.d.). ISEA. <https://www.infosecawareness.in/concept/student/internet-ethics>
- Services on the internet | Notes, videos, QA and tests | Grade 10>Computer Science>Internet and its services | Kullabs. (n.d.). KULLABS - An Award Winning E-Learning Platform Initiating Free Education. <https://kullabs.com/class-10/computer-science/internet-and-its-services/services-on-the-internet>

Chapter 3 – Digital Technology and Social Change

MODULE OVERVIEW

IT is an essential building block of the society. Presently, it has become the driving force behind social development. The module will hopefully provide a deeper understanding on how technological and social revolution go together.

This module will contain discussions on digital technology and its role in social change.

MODULE LEARNING OUTCOMES

At the end of this module, students are expected to:

- Explain the role of ICT in social change.
- Understand the strengths, weaknesses, opportunities, and imminent threats of the digital age.
- Articulate basic, but fundamental definitions of complex issues and dynamics that humans encounter every day, such as technology, social progress, development, and digitalization.

PRE-TEST 3

Answer the following questions to gauge your knowledge on the topic.

1. What social media sites do you use? How important are these to your day-to-day activities?
2. How does social media and societal issues intertwine?
3. Does social media promote or destroy societal progress? Why?

LEARNING CONTENT (INTRODUCTION TO THE DIGITAL AGE)

The **Digital Age** is simply the time wherein enormous amounts of information are made available to many people through digital computers. Initially, the underlying technology used in the digital age was invented in the later quarter of the 19th century.

▪ **Origins of the Digital Age (1947-1969)**

Transistors and integrated chips were created between the 1940s and the late 1960s. Moore's Law predicts the pace of rise in performance and size of integrated circuit chips, predicting that the number of transistors in an IC would double every two years. Based on this finding, it is projected that the manufacturing of integrated circuits for electronic devices would increase year after year. Moore's Law has been utilized as a roadmap for semiconductor and electronics businesses' digital breakthroughs.

These electronic devices led the way to more advanced digital computers.

▪ **Invention of the Internet and the Introduction of Personal Computers (1969-1989)**

Packet-switched networks were developed in the late 1960s to the early 1970s. This has brought the development of the **Internet**; wherein multiple networks are joined to make a network of networks.

The 1970s is also the time in which home computers were introduced. Home computers are more affordable than the computers used in business and in the academe (scientific- and engineering-oriented computers). This implies that even if a buyer lacks technological understanding, they may obtain inexpensive computers. Computer specialists anticipated at the time that digital technology has stealthily crept into our lives and households.

In this era, semi-ubiquity has made their way into homes, schools, and businesses. Automated teller machines and industrial robots were used in banking & finance and in fabrication companies; video games and electronic music became popular with the youth; mobile phones and digital cameras were used for communication and event-recording.

- **Invention of the World Wide Web (1989-2005)**

Tim-Berners Lee developed the World Wide Web in 1989. It became open to the public in 1991. The internet swiftly grew in popularity, and by 1999, every country had an internet connection.

In the 2000s, an increasing number of people owned cell phones. Text messaging became a cultural phenomenon at this time.

- **Social Media and Web 2.0 (2005-present)**

In 2005, the population of the Internet reached 1 billion; in 2020 67% of the world's population is connected to the internet. Cloud computing has also entered the mainstream, with hundreds of companies investing on cloud platform.

With the advent of smartphones come **social media**.

LEARNING CONTENT (THE ICT AND ITS ROLE IN SOCIAL CHANGE)

Since the products of ICT is all around us, it is intuitive that information technology influences our society. An **information society** has ICT as the main driver of rapid information growth. Through this information growth, there are subsequent social changes.

Social informatics focuses on teaching the social issues of computing. Theories in technology and social change include the following:

- **Social Construction of Technology (SCOT).** Advocates of SCOT theory believe that technology does not determine human action, rather, human action shapes technology.
- **Social Shaping of Technology (SST).** Technology is not neutral, according to this view, and is shaped by society's dominating social, political, and economic values. As

a result, shifts in values result in various technological outputs, and social science concerns can be utilized to shape technologies.

To add, according to Peter-Paul Verbeek, a professor of technological philosophy at the University of Twente in the Netherlands, technology already has an impact on our moral decision-making, which impacts human agency, privacy, and autonomy. He warns against considering technology solely as a tool for humans, instead advocating for it to be viewed as an active agent.

Social Media and Its Role in Social Change

Since the early 2000s, the usage of social media has grown dramatically. Social networking sites such as Facebook and Twitter, among others, were created in this time. Given the global diversity of these social media sites, it is difficult to generalize about their purpose, although they are focused on facilitating social interaction networking, such as linking friends. Nonetheless, the use of social media for societal change and political activity has the potential to be essential since these networks are built around key social communities and relationships that may be used to affect society.

Filipinos are declared as the most active users based on multiple web-based social media sites, and the Philippines has been dubbed as the “Social Media Capital of the World.” Social networking in the Philippines has extensive uses, including: (a) information dissemination; (b) promotion of e-commerce platforms; (c) as a form of electoral campaign, (d) as tools to aid criminal investigation, and more.

LEARNING CONTENT (GLOBAL DIGITAL DEVELOPMENT)

Research, development, and deployment of digital technologies are important in sustainable development.

Information and communication technologies (ICTs) are beneficial in the development of the society; its importance is highlighted during the COVID-19 crisis wherein ICTs are instrumental in keeping healthcare and economy afloat.

- **ICT Infrastructure for Digital Development**

The **International Telecommunication Union (ITU)** facilitates the improvement of telecommunication infrastructure in the developing world. During the ongoing pandemic, there is increased dependency on digital infrastructure and reliable internet access. As countries impose social distancing measures, it is important to have reliable internet access to go about our normal lives. ITU calls for the improvement of the digital services provided by telecommunications operators

- **MSMEs and ICTs**

Micro, small, and medium enterprises (MSMEs) make use of digital technology in adapting with the difficulties brought by the COVID-19 pandemic. Small businesses use **social media sites** to market their products; business owners also deal with **e-commerce**, which is the facilitations of transactions through online means. MSMEs are given a platform through digital technology.

ICT can impact the growth of MSMEs in our society.

- **Connecting Health Services for Public Welfare**

Through digital technology, we can connect health centers and hospitals to their patients. This enables health services such as **telemedicine** and remote diagnostics.

LEARNING CONTENT (TECHNOLOGICAL CHANGE)

Technological change (TC) refers to the total process of creating, innovating, and disseminating technologies or processes. technical advancement is based on the improvement and expansion of technology. Technological change includes the following:

1. **Invention of Technologies**

Finding new or **breakthrough** technology (including products and processes) is included in product development. When modern technologies are invented, they are patented. Utility patents are done for new processes, machinery, articles of manufacture, and others.

2. **Commercialization or open-source release through research and development (producing emerging technologies)**

Transitioning technologies from the research lab to the marketplace is called technology commercialization. Getting a patent or licensing an invention takes it into the market to create financial value. R&D agreements between science institutions and the industry provides research institutes with funding.

Meanwhile, open-source release allows a copyright holder to grant users the right to use, study, change and distribute their product. This occurs in open-source software. **Open-source software development** allows the collaboration of online users in the development of the software. Through open-source code, end users can change and adapt their software to their personal needs; this is done since the modular software allows for the development of independent components.

One of the prominent organizations that is involved in open-source software development is the Linux Foundation, which has spawned popular Linux distributions such as Debian, Ubuntu, and Fedora.

Commercialization and open-source release are both examples of technology transfer.

3. **Continuous technological improvement (often resulting in lower costs)**

Through technological innovation, previous versions of a product are used as motivation to create new or better products. Research and development take on the role of developing new services and improving existing ones. Patents and other scientific advancements are fueled by R&D, which leads to productive growth in industries including manufacturing, medicine, engineering, and government.

4. **Technology dissemination throughout industry or society**

Technology dissemination or diffusion is the spread of technology in the industry, and in the society. As an example, computers were used in business settings such as office

workstations; then, personal computers have made its way into our homes. Normally, the dissemination of technology involves disruption and convergence: these are periods of success and stagnancy in the adoption of technology.

LEARNING CONTENT (SOCIAL EVOLUTION)

Information technology has the potential to promote social empowerment and social transformation. IT has profoundly altered our information and communication paradigms by allowing multiple ways for communication and information flows that are easy and affordable. Another major paradigmatic characteristic of IT is the ability for asynchronous communication (i.e., emails in which both sender and receiver need not be connected at the same time) and even synchronous communication (through video conferencing, 5G communication, etc.). Furthermore, digital information processing may assist in different work and professions that were previously deemed unimaginable.

Here are some indicators on how ICT has empowered communities.

Indicators for Community Empowerment (through ICT use) - Social Capabilities strengthened **adopted from Gigler, Bjorn-Soren

Dimension	Objective	Outcome indicator
Informational	<i>To improve access to information and informational capabilities</i>	• <i>information system strengthened</i> • <i>information flow within community improved</i> • <i>horizontal and vertical knowledge exchanges with other communities</i>
Organizational	<i>To strengthen organizational capabilities</i>	• <i>transparent selection of leaders</i> • <i>increased efficiency</i> • <i>improved information flow and coordination among different organizations</i>
Social Development	<i>To improve access to basic social services</i>	• <i>improved access to formal and non- formal education (i.e., e-learning)</i> • <i>improved knowledge and access to health services</i> • <i>improved knowledge and access to social programs of the government (e-government services)</i>

Economic Development	<i>To promote economic opportunities</i>	• <i>improved access to markets and commercialization of products</i> • <i>improve productive activities through enhanced knowledge (i.e., better knowledge about agricultural practices)</i> • <i>enhanced capacity to mobilize resources from outside donors</i>
Political Participation	<i>To improve participation in the political system</i> <i>To enhance transparency within community</i>	• <i>improved 'voice' and participation in development process</i> • <i>improved transparency of political community institutions (e-government)</i> • <i>enhanced information-based decision-making through easier information processing</i> • <i>better coordination of political activities in the community</i> • <i>direct participation in government activities</i>
Cultural Identity	<i>To strengthen the communities' cultural identity</i>	• <i>mainstreamed appreciation of culture and tradition</i> • <i>improved dissemination of communities' own culture</i>

POST-TEST 3

Answer the following questions:

1. Based on the discussions, how do SCOT and SST differ?
2. List down the social media sites you use. What do you use it for? How does it affect your lives daily?
3. Technological change is done in multiple stages. Explain how computers have evolved through the stages of technological change.

SUMMARY

- Through information technology, enormous amounts of information are made available to many people through digital computers.
- Information technology influences our society. Two theories try to explain how IT and social change intertwine: the **Social Construct of Technology** and the **Social Shaping of Technology**.

- Social media has grown dramatically since the 2000s. Social networking sites make it easier to promote social and political campaigns, disseminate information, and other purposes; thus, it has a function in shaping social activity of its users and other individuals.
- Research and development (R&D) are made easier through information technology. This allows the advancement of IT infrastructure, business, and commerce, as well as healthcare systems all over the world.
- Technological change is the process of creating new inventions, marketing and innovation, and the dissemination of modern technology.
- IT can influence social transformation.

REFERENCES

BOOKS/E-BOOKS:

E-SOURCES:

- Gigler, Bjorn-Soren. (2004). Including the Excluded-Can ICTs empower poor communities? Towards an alternative evaluation framework based on the capability approach.
- Gleick, James (2011). The Information: A History, a Theory, a Flood
- Greenfield, P. (2019, August 8). Communication technologies and social transformation: Their impact on human development. DOI:10.1017/9781108264846.009
- Gutman, Dan (September 1987). "What happened to the computer revolution?". Commodore Magazine.
- Miladi, N. (2016, February). Social media and social change. https://www.researchgate.net/publication/294579134_Social_Media_and_Social_Change
- PH remains top in social media, internet usage worldwide – report. (2021, January 28). Rappler. <https://www.rappler.com/technology/internet-culture/hootsuite-we-are-social-2021-philippines-top-social-media-internet-usage>
- The role of social media in societal change: Cases in Finland of fifth estate activity on Facebook - Nina Sormanen, William H. Dutton, 2015. (2015, November 5). SAGE Journals. <https://journals.sagepub.com/doi/full/10.1177/2056305115612782>

Chapter 4 - Overview of Ethics

MODULE OVERVIEW

This module will discuss the overview of ethics by defining ethics, identifying values and attitude anchored to ethics, the different ethical principles, and its application, as well as the different forms of ethical theories.

MODULE LEARNING OBJECTIVES

At the end of this, students are expected to:

- Define ethics, values, and attitude
- Discuss the different ethical principles
- Differentiate the forms of ethical theories

PRE-TEST 4

Answer the following questions to gauge your knowledge on the topic.

1. How does ethics differ from one's values?
2. What ethical principles do you follow in your life?
3. How does ethics play in your role as a student in the university?

LEARNING CONTENTS (Ethics, Values, and Attitude)

What is Ethics?

The set of standards of right and wrong that govern human behavior, it involves recommending concepts of what humans ought to do. It can also refer to as a set of accepted beliefs of individual or group of people based on morals that control human behavior. Ethics can be seen in the form of codes of conduct implemented personally or by organizations. Ethics has something to do with human behavior and it have direct impact on human values and attitude. Values can contribute to determine appropriate standard behavior, and attitude is manifested in persons behavior.

Values

It can simply define as something that is worthy or important to an individual. Person's belief affects evaluative judgments about what is right or desirable. How personal character will develop, people think and behave determined by things that a person regard as good or right. Some examples of values are respect, honesty, personal responsibility, or kindness.

Attitude

It refers to person's strong belief or feeling towards people, things, and situation. Attitude can be positive or negative and it can influence the behavior of those around them. Example of positive and negative attitudes are confident, cooperative, considerate, cheerful, unfriendly, selfish, snobby and the like.

LEARNING CONTENTS (Ethical Principles)

Ethical Principles

It is the basis of ethical analysis and serves as the viewpoints in making decisions. These are the ethical recommendations and evaluations of human actions in justifying judgments. It determines the considerations that should be weighed in decision making or judgment. The different ethical principles are beneficence, least harm, respect for autonomy, and Justice.

Beneficence

It refers to an act of charity, mercy, and kindness with a strong connotation of doing good to others including moral obligation. A person's duty to act in the best interests and well-being of the patient, client, or resident. This action can be in a form of kindness, mercy, and/or charity with implication of doing good to others and moral obligation. However, doing more than is required or beyond what is obligatory is called exceptional beneficence or classified as supererogatory. For example, providing an extra benefit to clients that exceeds to the required services.

Least Harm (Non-maleficence)

A person's duty to do no harm, to protect others from harm, and to maintain one's professional competence. It is like beneficence except that it can be applied to situations in which neither choice is beneficial where a person should choose to do the least harm possible and to do harm to the fewest people. A person has a larger responsibility to simply walk past other person rather than to make derogatory remarks about that person.

Respect for Autonomy

A person's duty to respect and promote individuals' decision for themselves or apply to their lives in achieving what they believe to be in their best interests that includes respect for privacy and confidentiality. For example, a patient can choose or decide about their medical treatment.

Justice (Distributive Justice)

All persons involved, being of equal moral worth, should be treated fairly. Ethical decisions should be consistent with the ethical theory except there is a situation that provide an excuse and justification for an action. For example, A police officer is allowed for over speeding to arrive in the crime scene immediately and to prevent a person from getting hurt.

LEARNING CONTENTS (Forms of Ethical Theories)

Ethical Theories

Ethical theories are based on ethical principles that points up different aspects of an ethical conflict and serves as guide to ethically correct resolution that is not beyond the ethical theory guidelines. The choice of a person in ethical theory commonly based on their experiences. There are four categories of ethical theory that includes deontology, utilitarianism, rights, and virtue.

Deontology

A category of ethical theory stating that people should adhere to their obligations and duties when performing decision making in an ethical conflict. Upholding one's duty is considered ethically correct where a person follows his/her obligations to another individual or group of people. A deontologist produces consistent decisions since they will be based on individual's set of duties. The flaw of this theory is that there is no rationale or logical basis for deciding an individual's duties. The reasons why people chose to make it as their duty is unknown. Another flaw is the conflict on duties and concern with the welfare of others, for example a person's duty is to attend a meeting on time and the situation of this person is running late, how this person supposed to drive? Speeding breaks his/her duty in the society and on the other hand he/she will arrive at the meeting late that will not fulfill his/her duty.

Utilitarianism

It is founded on one's ability to predict the consequences of an action. For utilitarian, the ethically correct choice is the option that yields the greatest benefit to the most people. It compares predicted solutions and determine the choice which is more beneficial for more people.

Two Types of Utilitarianism:

1. *Act Utilitarianism* – conforms to the definition of utilitarianism, which sticks to actions that benefit the greatest number of people regardless of personal sentiments or society restraints such as laws.
2. *Rule Utilitarianism* – actions are concerned with fairness and takes law into account. It seeks to benefit the most people however through the fairest and most just means available. It values justice as well as beneficence at the same time.

Act and rule utilitarian have disadvantages although life experience can be applied in predicting outcomes, it is not certain that these predictions are accurate. This flaw can lead to unexpected results and the decision maker appear unethical. Another issue of utilitarian decision makers is the ability to assess different sorts of outcomes on a similar scale. It is also difficult to compare tangible gains to intangible gains (such as money and happiness) since their attributes differ so drastically.

Achieving the maximum good is the concern of act utilitarian. Consequently, one individual's right may break the terms to benefit a greater number of people. This type of utilitarianism is not always concerned with autonomy, beneficence, and/or justice for an individual if persecuting leads to a solution that benefits majority of people. Another challenge for act utilitarianism occurs when an individual faces sudden changes in experiences in the set of variable conditions that can lead to changes in the decision.

The possibility of conflicting rules occurs in rule utilitarianism. It may encounter conflicting ideas about what is ethically correct that leads to no ethically correct answer. For example, the scenario of running late in the meeting. An individual may believe that following the law would benefit society and at same time he/she may also believes that arriving on time in the meeting is ethically correct since it also benefits the society.

Rights

The ethical theory that based on rights are the rights that established by a society and are given the highest priority and protected. When the rights were endorsed by a large population,

it is considered valid and ethically correct. Rights can also be grant to others if they have the ability and resources.

On a larger scale, this theory has major complication. Understanding the characteristic of a right in the society is a must and the society must determine what rights it wants to sustain and give to its citizens. To be able to do that, they must decide what is the ethical priorities are and the society's goals. Hence, to make the rights theory useful, it must be used concurrently with another ethical theory that explain the goals of the society constantly. For example, upholding the right to freedom of religion was one of the goals of a society, and this right was upheld in the Constitution then people have the right to choose their religion.

Virtue

The virtue ethical theory judges the character of a person rather than an action that may veer from the person's normal behavior. When rating an unusual behavior that considered unethical, they must consider the person's reputation, morals, and motivation.

For example, if a person plagiarized a part of a paper that was subsequently discovered by a peer, the peer will grasp the person's character and will criticize the person accordingly. If the plagiarizer typically respects the rules and is well-liked by his peers, the peer who comes across the plagiarize paper may be able to assess his friend more leniently. Perhaps the researcher had a late night and just forgot to properly cite his or her source. A person with a reputation for academic misconduct, on the other hand, is more likely to be evaluated harshly for plagiarizing due to a persistent history of unethical activity. The change in a person's moral character is not taken into account, which is a drawback of this form of ethical theory.

POST-TEST 4

1. What is the impact of ethics in a person's values and attitude?
2. Give example or scenario that may apply in each ethical principle.
3. Differentiate the forms of ethical theory.

SUMMARY

- Ethics is a set of standards of what is right and wrong used to govern human behavior; it entails advocating conceptions of what individuals should do. It is related to human conduct and has a direct influence on human values and attitudes.
- Values is something that is worthy or important to an individual.
- Attitude refers to person's strong belief or feeling towards people, things, and situation.
- Ethical principles are ethical suggestions and assessments of human acts used to explain judgements. Beneficence, least damage, autonomy, and justice are the many ethical principles.
- Ethical theories are founded on ethical principles that highlight various elements of an ethical problem and serve as a guide to morally right resolution that does not go beyond the ethical theory parameters. Deontology, utilitarianism, rights, and virtue are the four categories of ethical philosophy.

REFERENCES

E-SOURCES:

- *Personal values and ethics – Organizational behavior*. (2019, February 27). BCcampus Open Publishing – Open Textbooks. Adapted and Created by BC Faculty. <https://opentextbc.ca/organizationalbehavioropenstax/chapter/personal-values-and-ethics>
- Principles and theories. (n.d.). Biology at Davidson. <https://bio.davidson.edu/people/kabernd/indep/carainbow/Theories.htm> [EthicalTheories. pdf \(dsef.org\)](#)
- *The principle of beneficence in applied ethics (Stanford encyclopedia of philosophy)*. (n.d.). <https://plato.stanford.edu/entries/principle-beneficence/>
- What is ethical principles - Meaning and definition - Pallipedia. (n.d.). Pallipedia - The Free Online Palliativ
- Care Dictionary. <https://pallipedia.org/ethical-principles/>

Chapter 5 - Ethical Issues in Digital Age

MODULE OVERVIEW

As data become available over the internet, ethical issues arise as well. Digital age makes our data vulnerable to different types of ethical issues. This module will discuss the privacy, privacy invasion, hacking, theft, copyright infringement, freedom of expression and online defamation, piracy, fraud, and Internet pornography.

At the end of this, students are expected to:

- Identify the different ethical issues in digital age.
- Describe the threats posed by hackers, fraud, piracy, and internet pornography
- Identify the different types of theft
- Discuss the copyright infringement
- Discuss the freedom of expression and online defamation

PRE-TEST 5

Answer the following questions to gauge your knowledge on the topic.

1. What are some ethical issues that you know of? How does the use of technology affect the spread of these ethical issues?
2. What is the difference between libel and defamation?
3. How does one draw the line in freedom of expression and in libel/defamation?

LEARNING CONTENTS (Ethical Issues in Digital Age)

1. Privacy

According to law, privacy is the right of every individual to make personal decisions and conduct their life in a reasonable manner away from public observation. In other terms, it is an individual's or a group's capacity to choose express themselves and keep themselves or information about themselves private.

Privacy Invasion

It is a legal term that describes a circumstance or condition in which a person intrudes on the privacy of others, such as the abuse or disclosure of sensitive information and unlawful monitoring. Invasion of privacy is an unlawful conduct that carries legal ramifications since it causes loss or hurt to people's life.

Types of Privacy Invasion

- a) *Intrusion into seclusion* – It refers to the purposeful intrusion of another person's private affairs. It also involves the act of looking for private information of a confidential nature, even if the information is not publicly available. In this sort of tort, the legal criterion should be deliberate and extremely offensive to a reasonable person.
- b) *Appropriation of name or likeness* – it refers to the act of using the name and likeness of other person publicly without authorization for his/her own benefit such as trading and advertising purposes.

- c) *Public disclosure of private facts* – it refers to the publication of other person's private affairs with the state that it is highly offensive to a reasonable person and in legitimate public concern.
- d) *Placing a person in a false light* – act of stating information that is based on opinion and may be technically true, but indicates to a reasonable person that anything bad and/or incorrect about the target is true. It is similar to defamation, except that no false statements are required.

2. Hacking

System penetration is the act of getting unauthorized access to data via a system or other digital device such as a computer. Because information in digital form is subject to numerous forms of assaults, it is one of the most prevalent dangers in cybersecurity. There are various labels for different types of hacking, which include the following:

- a) *Black-hat hacker* – it refers to an act that takes advantage of security vulnerabilities to gain unlawful access to private networks for the purpose of private advantage.
- b) *White-hat hacker* – it is a type of hacker that works for the common good by hacking networks to call attention to flaws in security to be fixed.
- c) *Gray-hat hacker* – It refers to the person who enact activities of both black hat and white hat hacker.
- d) *Script Kiddie* – a novice in hacking. To hack networks, it follows instructions and use pre-existing software.

There are also different tools and methods used in hacking:

- a) *Key-logging software* – it is remotely installed on a computer to record keystrokes and commands. These recordings will be collected over the Internet and played back to spy and steal username and password.
- b) *Packet-sniffing software* – it is used to steal private data that is being transported over a network. It focuses on capturing and analyzing packets that flows over the network or Internet. It might capture important information such as passwords and credit card numbers.
- c) *Port-scanning software* – it is used to search random IP addresses for open ports to be hacked.
- d) *Social Engineering* – it refers to the exploitation of natural human tendency to trust others. For example, pretending to be an administrator through phone call to make the person to provide a password or other important information.
- e) *Dumpster diving* – a method of going through trash of other user to steal personal information such as credit card numbers.

3. Theft

Another ethical issue concerning the use of computer is theft. The most popular type of theft is known as identity theft – an act of stealing other's personal information to commit fraud. It can also be in the form of pretending to be someone else, using the information of other people for money and other benefits. An identity thief can make unauthorized purchases and transactions using other person's credit card information.

Personal information can be obtained using computer technology by accessing computer-based public records, hard drives of discarded and stolen computers, hacking computer

networks using information-gathering software/malware, or by simply sending deceptive text messages or emails, and browsing social networking sites.

4. Copyright Infringement

The act of using the copyrighted works or covered by copyright law without permission. Specifically, reproducing, distributing, displaying, performing protected work, or making derivative works without authorization of the copyright holder. A copyright holder can be a creator of the work, publisher, or other business that copyright was assigned where they invoke legal and technological measures to prevent and penalize copyright infringement.

Copyright infringement in the Philippines constitutes the following statement: "Under Philippine law, copyright infringement occurs when there is a violation of any of the exclusive economic or moral rights granted to the copyright owner. It may also consist in aiding or abetting such infringement. The IP Code also provides for the liability of a person who at the time when copyright subsists in a work has in his possession an article which he knows, or ought to know, to be an infringing copy of the work for the following purposes: (a) selling or letting for hire, or by way of trade offering or exposing for sale or hire, the article; (b) distributing the article for the purpose of trade, or for any other purpose to an extent that will prejudice the rights of the copyright owner in the work; or (c) trade exhibit of the article in public."

Different forms of copyright infringement:

- Film recording in a movie theater
- Posting a video on your company's website that features copyrighted contents such as songs, wordings, and images
- Modifying copyrighted image or videos and displaying it to company's website
- Creating merchandise for sale that features copyrighted works
- Downloading films or music without paying their use
- Copying any artistic or literary work without license

5. Freedom of Expression and Online Defamation

Freedom of expression is the right to search, receive, and impart information of your choice regardless of boundary without interference. It is the right of every person to express themselves even online and to access information, opinions, and expressions of others including views on religion, political speech, opinions, and expressions that are inoffensive, but also expressions and opinions that might be offensive and disturbing to others. Having the right to express yourself should consider the rights of others and right to privacy.

Online Defamation is the act of giving false statement that is harmful to someone's reputation verbally or in written form using online platforms such as social media platform. There are two types of defamation:

- a) *Libel* – it refers to the written defamatory statement
- b) *Slander* – it refers to the spoken defamatory statement

For example, if the defamatory statement is posted online such as comment in a Facebook

post, it is considered as libel and identified as online defamation. Other example includes public comments on media websites such as online newspapers or magazine, chatrooms, and blog posting comments.

Is freedom of expression can be a defamation?

Statements that are labeled as merely opinions does not make it as defamation. Courts focuses on statement of verifiable facts which means the statement conveys probable false factual assertion such as cheating on his/her spouse, or someone committed murder.

6. Piracy

Privacy entails the unauthorized reproduction, use, and distribution of digital intellectual property such as software, music, and movies. Piracy and plagiarism are types of violations of intellectual property laws, such as copyright offenses. Pirated software, music, and films can be obtained via file sharing networks or handmade CDs and DVDs.

Examples of Piracy includes the following:

- Illegal use or interception of internet connection of other people
- Publication, reproduction, and/or use of patented or copyrighted work without permission
- Unauthorized copying of protected software
- Using protected software without license

7. Fraud

In general, it is a wrongful act and criminal deception resulting in financial or personal gain. *Computer fraud* is the term used in executing the illegal act using computer system or attacking a computer intends to alter, damage, or worst disable it.

Types of fraud includes:

- a) *Mail Fraud* – fraudulent activity that uses mails or emails such as sending a letter to someone to scam money or gain personal information. It also includes using chain letters to collect items or money, or by simply stealing and opening someone else's mail.
- b) *Health Care Fraud* – it refers to a situation where an individual such as insurance provider or medical office misuse the insurance information for their own personal gain.
- c) *Debit and Credit Card Fraud* – accessing other people's debit or credit cards that can completely drain bank accounts.
- d) *Internet Fraud* – fraudulent activity using internet to take advantage to other people. It includes data breaches, malware, and phishing.

8. Internet Pornography

Accessing pictures or films that contains representation of sexual behavior using Internet. It is accessible over the Internet via file sharing and websites. The increasing availability of public access via the World Wide Web fuels the proliferation of internet pornography. Pornography is illegal for children, yet it has grown so accessible and widely available that it is seen as one of the most serious hazards to children's internet safety.

Internet pornography is regarded as illegal when there are minor children involved. Online child pornography has become a major issue, and it can take the form of pictures, movies, drawings, animations, and even computer-generated imagery. In certain circumstances, child pornography involves an adult being portrayed as a kid. Child pornography can also entail the alteration of a picture to make it appear like a child.

Internet pornography includes uploading and downloading pornographic materials via internet illegally, this is also regarded as internet pornography crime.

POST-TEST 5

1. Describe the positive and negative impacts of technology on freedom of expression.
2. Watch the following video: <https://www.youtube.com/watch?v=PYhNWpXWLLc>. Based on the video clip, what are the laws implemented for health records? How do these laws protect our health records?
3. How can you protect your personal information against identity theft?

SUMMARY

- Privacy invasion is a legal phrase that describes a circumstance or condition in which a person intrudes on the privacy of another, such as the abuse or disclosure of personal information or unlawful monitoring.
- Hacking is the act of getting illegal access to data through the use of a system or other digital equipment, such as a computer, and is also known as system penetration.
- Theft is an act of stealing other's personal information to commit fraud.
- Copyright infringement is the act of using the copyrighted works or covered by copyright law without permission.
- Freedom of expression is the right to search, receive, and impart information of your choice regardless of boundary without interference.
- Online Defamation is the act of giving false statement that is harmful to someone's reputation verbally or in written form using online platforms such as social media platform.
- Piracy involves illegal copying, use, and distribution of digital intellectual property such as software, music, and movies.
- Fraud is a wrongful act and criminal deception resulting in financial or personal gain.
- Internet pornography refers to accessing pictures or films that contains representation of sexual behavior using Internet.

REFERENCES

BOOKS/E-BOOKS:

- Stair R., Baldauf K. *The World of Information Technology*. 2009. Cengage Learning.

E-SOURCES:

- *Copyright law in the Philippines* | Federis. (n.d.). Federis & Associates Law.

- <https://www.federislaw.com.ph/faqs-resources/copyright/>
- *Freedom of expression and information.* (n.d.). Freedom of Expression. <https://www.coe.int/en/web/freedom-expression/freedom-of-expression-and-information>
 - *How to avoid copyright infringement.* (2015, December 2). LegalZoom: Start a Business, Protect Your Family: LLC, Incorporate, Wills, Trademark, Legal Advice. <https://www.legalzoom.com/articles/how-to-avoid-copyright-infringement>
 - *Internet pornography crimes.* (n.d.). Jackson Personal Injury Law Firm | Coxwell & Associates. <https://www.coxwelllaw.com/internet-pornography-crimes.html>
 - Privacy legal definition of privacy (thefreedictionary.com). <https://legal-dictionary.thefreedictionary.com/privacy>
 - *The torts of invasion of privacy.* (n.d.). Lawshelf Educational Media. <https://lawshelf.com/shortvideoscontentview/the-torts-of-invasion-of-privacy/>

Chapter 6 – Privacy

MODULE OVERVIEW

Privacy can be described as imposing constraints on access to and/or use of personal information.

Privacy refers to the usage of information collected about individuals and stored in databases. This module will discuss the difference between privacy and security, explain various risks to Internet privacy, and identify and understand the different key privacy and anonymity issues.

MODULE LEARNING OBJECTIVES

At the end of this, students are expected to:

- Difference between privacy and security.
- Explain various risks to Internet privacy.
- Identify and understand the different key privacy and anonymity issues.

PRE-TEST 6

Answer the following questions to gauge your knowledge on the topic.

1. Do you share your personal information online? How and how often do you share these information?
2. How can technology affect one's privacy and anonymity?
3. What are some laws in the Philippines (and in other countries/organizations) that promotes one's data privacy?

LEARNING CONTENTS (Privacy Protection and the Law)

WHAT IS PRIVACY?

Privacy focuses on having control and regulation over the sharing or revealing of certain data about yourself or your behavior. Privacy is often hard to return by lately because we are so connected and open with our information. Two decades ago, internet privacy would have meant ensuring your computer's network connection was secure. Now a days, we use more internet-connected devices during a single day, including your smartphone, smart home device, smartwatch, tablet, and laptop.

Much of the web today is being employed by big companies like Google, Facebook, and Amazon trying to gather people's data for monetization and marketing purposes. Another problem experience is that not only our privacy being captured, but it's also being given in a privacy-focused products. In many cases, when we're given access to a free product, we pay with our data information. If we don't want to be tracked, we often need to prefer to use paid, privacy-friendly options.

INFORMATION PRIVACY

Information privacy is that the relationship among the gathering and circulation of information, technology, the public expectation of privacy, and therefore the legal and political issues surrounding them. It is also best-known as data privacy or data protection.

Data privacy is difficult since it try to use data while securing a person's privacy preferences and personally typical information. The fields of computer security, data security, and knowledge security all design and use software, hardware, and human resources to deal with this issue.

Information privacy, data privacy or data protection laws render a legal structure on the way to get, use and store data of natural persons. The various laws all over the world describe the interest of natural persons to regulate who is using its data. This includes usually the proper to urge details on which data is stored, for what intention and to request the deletion just in case the aim isn't given anymore.

- ✓ For all data gathered, there should be a stated purpose.
- ✓ Information collected from a private can't be expose to other organizations or individuals unless clearly authorized by law or by consent of the individual.
- ✓ Records kept on an individual should be authentic and up to date.
- ✓ There should be instrument for individuals to review data about them, to check accuracy. This may include periodic reporting.
- ✓ Data should be removed when it's not needed for the stated purpose.
- ✓ Transmission of private information to determine where "equivalent" personal data protection can't be confirmed is prohibited.
- ✓ Some data is just too sensitive to be collected, unless there are severe circumstances (e.g., sexual orientation, religion).

What is Privacy Protection?

Privacy protection is keeping the information you'd like to secure to yourself from getting into the custody of companies, hackers, government organizations, and some other groups. The explanation of privacy protection changes from person to person. Each person has various assumptions of privacy, therefore the level of security they want to feel that their privacy is really secure limit importantly.

Advantages of Privacy Protection

- **Protect Your Personal Data**

Privacy protection keeps your personal data secure from people that might want to take advantage of it. Minimizing your digital trace makes it harder for people to make advantage of your data.

- **Stop Unwanted Solicitations**

Without privacy, marketers can target advertisements and other messaging at once to you. If these sorts of ads bother you, the simplest way to avoid them is to stop the marketers from getting your data. Privacy-intrusive data collection runs the web advertising industry. If you'll end the flow of information you give

to the marketers, they'll not be ready to target you with ads.

- **Protect Your Email Address**

If you'll keep third-parties from acquiring your email address, you'll avoid getting plenty of unwanted spam email. Willingly giving out your email leads to an intense flow of advertising messages in your inbox.

Disadvantages to Privacy Protection

- **Privacy-Focused Products Cost Money**

Keeping your personally distinctive data secure isn't free. Because more mainstream services gather and exchange your data to form money, private alternatives need to adopt various business models. Free tools often collect your information, so you don't need to pay. Privacy-friendly business models often use a subscription-based payment model.

- **Is it Really Private?**

You can take all recommended step towards privacy and as to one deficiency, leak your information. It's our viewpoint that any privacy protection is best than none, but it could be overwhelming. If you sense like your efforts aren't being honor, it might be difficult to continue using privacy-focused products and services.

- **Your Information Might Still Be Public**

If you need privacy products your data should be private. Forgetting only one aspect of protecting your privacy could let your personal information slip into an enormous database and within the hands of an online advertiser.

- **Your Reputation May Be In Question**

There are some spots surrounding privacy on the web. Some people feel that the merely reason that somebody would want to hide their tracks online is because they're performing something illegal. While this isn't the case, it's an hopeless condition for anyone who sees the benefits of keeping your internet activity hidden from prying eyes.

PHILIPPINES – THE DATA PRIVACY ACT OR RA NO. 10173

In 2012, the Philippines passed Republic Act No. 10173 or the Data Privacy Act of 2012 (DPA) *“to protect the fundamental human right to privacy of communication while ensuring free flow of information to promote innovation and growth. The State recognizes the vital role of information and communications technology in nation-building and its inherent obligation to ensure that personal information in information and communications systems in government and in the private sector are secured and protected”*.

The DPA was passed in accordance with the Philippines agreements under ASEAN Vision 2020 and at the urging of the growing business process outsourcing industry. The law was

modeled after the Data Protection Directive (95/46/EC) with many of its terminologies and provisions like privacy laws in other jurisdictions.

What acts are covered by the DPA?

The DPA and its [Implementing Rules and Regulations](#) (IRR) apply to all acts done or practices engaged in and outside of the Philippines if:

- If the person, either an individual or an institution, involved in the processing of personal data is in the Philippines;
- The act or practice involves personal data of a Philippine citizen or Philippine resident;
- The processing of personal data is done in the Philippines; or
- The act, practice or processing of personal data is done by an entity with links to the Philippines, subject to international law and comity.

“Personal data” refers to all types of personal information.

“Processing” is any operation/s performed upon personal data. These operations include, but are not limited to the **collection, recording, organization, storage, updating or modification, retrieval, consultation, use, consolidation, blocking, erasure, or destruction of data.**

Who implements the DPA?

The National Privacy Commission (NPC) oversees administering and implementing the DPA. It is also tasked to monitor and ensure compliance of the Philippines with international standards for personal data protection. The major functions of the NPC are as follows:

- *Rulemaking.*
- *Advisory.* The NPC is the advisory body on matters related to personal data protection.
- *Public education.* The NPC shall launch initiatives to educate the public about data privacy, data protection and fair information rights and responsibilities.
- **Compliance and monitoring.** The body has compliance and monitoring functions to ensure personal information controllers comply with the law. It is also tasked to manage the registration of personal data processing systems.
- **Complaints and investigations.**
- **Enforcement.**

“Personal information controller” is an individual or institution, or any other body who controls the processing of personal data, or instructs another to process personal data on its behalf.

How to comply with the Data Privacy Act?

If you are a personal information controller, you must follow the following by the law:

Registration of data processing systems (DPS). An individual or institution employing fewer

than 250 employees need not register unless its data processing operations: involves sensitive personal information of at least 1,000 individuals; likely to pose a risk to the rights and freedoms of data subjects; or the processing is not occasional.

Notification of automated processing operations where the processing becomes the sole basis of making decisions about a data subject and when the decisions would significantly affect the data subject. A “*data subject*” is an individual whose personal, sensitive personal or privileged information is process.

NOTE: No decision with legal effects concerning a data subject shall be made solely based on automated processing without the consent of the data subject. The consent may be in written, electronic or recorded form. It may be given by a lawful representative or agent.

Appointment of a **Data Protection Officer** in charge of ensuring compliance with the DPA; **Creation of a data breach response team** that will at once address security incidents or personal data breach; **Adoption of data protection policies** that support data security measures and security incident management; **Annual report** of the summary of documented security incidents and personal data breaches; and **Compliance with other requirements** as may be provided by the NPC.

LEARNING CONTENTS (Key Privacy and Anonymity Issues)

PRIVACY AND ANONYMITY ISSUES

Data Breaches

Data breaches involving big databases of personal information have resulted in a tremendous number of identity theft occurrences. Data breaches are occasionally the result of hackers hacking into a database, but often, they are the result of negligence or a failure to follow adequate security standards.

Organizations are hesitant to reveal data breaches for fear of negative publicity and consumer litigation. Victims whose personal information was compromised because of a data breach, on the other hand, need to be informed so that they can take precautionary steps.

The cost of a data breach to a business is sometimes expensive. Approximately half of the value is often a result of missed business potential relating to clients whose patronage has been lost because of the occurrence. Other expenses include public-relations charges to protect the firm's reputation, as well as higher customer-service fees for information hotlines and credit monitoring services for victims.

Electronic Discovery

The compilation, preparation, examination, and production of electronically stored material for use in criminal and civil actions and procedures is known as electronic discovery (e-discovery). Electronically stored information (ESI) is any type of digital data, such as electronic mails, drawings, graphs, Web sites, pictures, word-processing files, sound/audio recordings, and databases, that is saved on any type of electronic storage device, such as hard drives, CDs,

and flash drives. Several types of ESI of a private or personal nature (e.g., personal emails) are extremely likely to be exposed throughout the e-discovery process.

Traditional software increase firms as well as legal organizations have established the growing need for enhanced processes to speed up and lessen the costs associated with e-discovery. As a result, lots of companies offer e- discovery software that cater the ability to do the following:

- Evaluate large volumes of ESI instantly to perform early case assessments.
- Simplify and assign data collection from across all relevant data sources in various data formats.
- Choose large amounts of ESI to lessen the include documents that need to processed and reviewed.
- Identify all participants in an investigation to figure out who knew what and when.

E-discovery increase many ethical issues: Should an organization ever plan to destroy or hide incriminating evidence that would rather be revealed during discovery? To what degree must an organization be proactive and careful in supplying evidence needed through the discovery process? Should an organization plan to hide incriminating evidence during a mountain of trivial, routine ESI?

Consumer Profiling

When people register on websites, take surveys, fill out forms, or take part in online competitions, companies fully collect personal information about them. Various firms also collect information on Web surfers using cookies, which are text files that are transferred to the hard disks of users who visit a website so that the website can recognize visitors on later visits.

Tracking software is frequently used by businesses to allow their websites to examine surfing behavior and deduce personal interests and preferences. The use of cookies and tracking software is dubious since corporations may obtain information about customers without their knowledge. Cookies, once saved on your computer, allow a website to customize the advertising and promotions that are displayed to you. The marketer is aware of which advertisements are being seen the least and ensures that they are not shown again unless the advertiser has chosen to promote utilizing repetition. Some cookies may also monitor which other websites a person has visited, allowing advertisers to use that information to make informed estimates about the types of advertising that would be almost fascinating to the user.

Consumer data privacy has evolved into a key marketing concern. Companies that are unable to safeguard or respect consumer information frequently lose business, and some become defendants in class action lawsuits arising from privacy violations.

Consumer profiling opponents are also worried that confidential information is being gathered and shared with other parties without the consent of the customers who supply the information. Consumers have no means of knowing how or who is using the information once it has been gathered.

Workplace Monitoring

There is enough data to support the idea that many workers squander significant amounts of time doing non-work-related activities. According to one recent survey, between 60% and 80% of workers' internet time is spent on non-work-related activities. According to another research, workers spend roughly four or five hours each week on personal problems on average. According to a recent poll conducted by an IT staffing business, 54% of firms reported restricting the usage of social networking to help reduce waste at work.

The actual for decreased productivity and increased legal liabilities has led many employers to watch workers to make sure that corporate IT usage policies are being followed. Many firms find it necessary to record and review employee communications and activities on the job, along with phone calls, email, and Web surfing. Some are even videotaping employees on the work. In addition, some companies use random drug testing and psychological testing. With some exceptions, these progressively common (and many would say intrusive) practices are perfectly legal.

Advanced Surveillance Technology

Several breakthroughs in information technology, such as surveillance cameras and satellite-based systems that can detect a person's actual position, provide incredible new data-gathering capabilities. However, these advancements have the potential to reduce individual privacy and simplify the problem of how much information about people's private lives should be kept protected.

- **Camera Surveillance**

Surveillance cameras are employed in major cities across the world to deter crime and terrorist activity. Critics claim that such inspection violates civil rights and are concerned about the cost of the equipment and additional persons needed to view the video feeds. Supporters of surveillance cameras provide report data indicating that the cameras are effective in preventing crime and terrorism. They can provide examples of how cameras aided in the resolution of crimes by verifying witness testimony and aiding in the identification of criminals.

- **Vehicle Event Data Recorders**

A vehicle event data recorder (EDR) is a device that collects vehicle and occupant data for a few seconds before, during, and after any vehicle incident severe enough to cause the air bags to deploy. Sensors located throughout the vehicle collect and record data such as vehicle speed and acceleration, seat belt use, air bag formation, activation of any automated collision notification system, and driver actions such as brake, accelerator, and turn signal usage. The EDR cannot collect any information that might be used to identify the vehicle's driver. It also cannot identify if the motorist was under the influence of drugs or alcohol while operating the car.

The fact that most automobiles now come equipped with an EDR, and that the

information from this device might be used as evidence in a court of law, is unknown to the public. The future capability of EDRs, as well as the extent to which their data will be used in court proceedings, remains to be seen.

- **Stalking Apps**

Technology has made it possible for one individual to track the locations of another person in the shortest amount of time possible, without ever having to follow the person. Cellphone spy software, also known as a stalking application, is frequently downloaded and installed on someone's cellphone or smartphone in minutes, allowing the user to perform location tracking, record calls, view every text message or picture sent or received, and record the URLs of any internet site visited on the phone. Even when the phone is switched off, the built-in microphone is frequently triggered remotely to be used as a listening device. All information gathered by such applications is frequently transferred to the user's email account, where it may be read in real time or at a later date. Some of the most popular spy software consist of Mobile Spy, ePhone Tracker, Flexi SPY, and Mobile Nanny.

There is no law that prevents a company from developing an app whose primary aim is to let one person track another, and this software is available for purchase on the internet. (Some users have said that they got virus after installing stalker applications or that the program did not work as described.) However, it is unlawful to install the software on a phone without the owner's permission. It is also unlawful to listen in on another person's phone calls without their knowledge and consent. These legal nuances, however, are not a deterrent to a persistent stalker.

POST-TEST 6

1. How much do you value your privacy?
2. Why does having knowledge about Privacy is so important?
3. Do you think the web increases privacy or takes away privacy?
4. What is the best idea to protect a personal privacy while not disrupt future innovation?
5. How is privacy perceive in your culture?

SUMMARY

- **Privacy** is the ability to regulate the sharing or exposing of certain facts about oneself or one's activity. Because we are so connected and open with our information, privacy is often difficult to regain.
- The link between the collecting and dissemination of information, technology, and the public expectation of privacy is referred to as **information privacy**.
- **Data privacy** is challenging since it aims to use data while safeguarding a person's privacy choices and personally identifiable information.
- **Privacy protection** is the process of preventing information that you want to keep private from falling into the hands of corporations, hackers, government agencies, and other organizations.

REFERENCES

BOOKS/E-BOOKS:

- Reynolds, G.W.(2015). Ethics in Information Technology Fifth Edition. Boston: Cengage Learning.
- Lavina, C.G. (2015). Social, Ethical, Legal and Professional Issues in Computing with complete explanation of the Philippine. Manila: Cybercrime Laws.Mindshapers Co., Inc.

E-SOURCES:

- *Definition of privacy*. (n.d.). Dictionary by Merriam-Webster: America's most-trusted online dictionary. <https://www.merriam-webster.com/dictionary/privacy>
- *Know your rights*. (2018, September 25). National Privacy Commission. <https://www.privacy.gov.ph/know-your-rights/>
- (n.d.). notitle. <https://www.officialgazette.gov.ph/>
- *Privacy and human rights - Overview*. (n.d.). Global Internet Liberty Campaign. <https://gilc.org/privacy/survey/intro.html>
- *Privacy and information technology (Stanford encyclopedia of philosophy)*. (n.d.). Stanford Encyclopedia of Philosophy. <https://plato.stanford.edu/entries/it-privacy/#ImplnfTecPri>
- *Privacy English definition and meaning | Lexico.com*. (n.d.). Lexico Dictionaries | English. <https://www.lexico.com/en/definition/privacy>
- *A summary of RA No. 10173 or the data Privacy Act of 2012*. (2020, January 16). ECC International. <https://eccinternational.com/ra-10173-data-privacy-summary/>
- *What is privacy protection? [Updated for 2021]*. (2021, March 25). Choose To Encrypt. <https://choosetoencrypt.com/privacy/what-is-privacy-protection/>

Chapter 7 - Freedom of Expression

MODULE OVERVIEW

Everyone has their right to freedom of speech and expression. This right includes freedom to express his opinions, and to share and receive information and ideas without interference from public authority.

This module discusses terminologies and concepts about one's freedom of speech and expression. It aims to equip the students with relevant knowledge to determine whether their claim to the right to speak and express is within the scope guaranteed by the Constitution.

This module will contain discussions on freedom of expression and how the internet has changed the way individuals share and receive information.

MODULE LEARNING OUTCOMES

At the end of this module, students are expected to:

- Defend their rights to freedom of expression
- Understand the limitation underlying the freedom of expression
- Determine the existing laws and acts concerning freedom of expression

PRE-TEST 7

Answer the following questions to gauge your knowledge on the topic.

1. What is freedom of expression?
2. Are there any limitations in one's freedom of expression? What existing laws define these limitations?
3. Does technology promote or limit one's freedom of expression? Why?

LEARNING CONTENT (Scope of Freedom of Expression)

What is freedom of expression?

Freedom of expression is a fundamental human right; this is protected by international law and every constitutional bill of rights in the world. It is essential for human growth, dignity, personal fulfillment, and the pursuit of truth. It is also a necessary requirement for democracy and effective administration of governing bodies. Freedom of expression allows for open debate about and between opposing political parties, allows individuals to express their concerns to authorities, and guarantees that new policies and laws are scrutinized.

Free speech improves the quality of government because it helps to ensure that officials are competent and honest, and it allows citizens to express their concerns about and discuss government action. To put it another way, democratic principles are jeopardized when information and ideas are restricted to flow freely. As a result, there is freedom to express one's views without fear of censorship, reprisal, or legal repercussions.

Freedom of speech, free speech, and freedom of expression are often used interchangeably. In a legal manner, freedom of expression includes any activity that imparts and seeks ideas, regardless of the medium used (peaceful assemblies, rallies, etc.).

Article 19 of the Universal Declaration of Human Rights states that:

Everyone has the right to freedom of opinion and expression; this right includes freedom to hold opinions without interference and to seek, receive, and impart information and ideas through any media regardless of frontiers.

The following statements from the European Convention on Human Rights (ECHR) focus on freedom of expression (Article 10 – Freedom of Expression):

1. Everyone has the right to freedom of expression. This right shall include freedom to hold opinions and to receive and impart information and ideas without interference by public authority and regardless of frontiers. This Article shall not prevent States from requiring the licensing of broadcasting, television, or cinema enterprises.
2. The exercise of these freedoms, since it carries with it duties and responsibilities, may be subject to such formalities, conditions, restrictions, or penalties as are prescribed by law and are necessary in a democratic society, in the interests of national security, territorial integrity or public safety, for the prevention of disorder or crime, for the protection of health or morals, for the protection of the reputation or rights of others, for preventing the disclosure of information received in confidence, or for maintaining the authority and impartiality of the judiciary.

Article 10 of the ECHR also enumerate the three components of the right to freedom of expression: (1) freedom to hold opinions; (2) freedom to receive information and ideas; and (3) freedom to impart information and ideas. Meanwhile, The Philippine Constitution provides the following notions on freedom of expression:

Article III Section 4. “No law shall be passed abridging the freedom of speech, of expression, or of the press, or the right of the people peaceably to assemble and petition the government for redress of grievances.”

Article III Section 18. “No person shall be detained solely by reason of his political beliefs and aspirations.”

Limitations of Freedom of Expression

The exercise of freedom of expression includes duties and responsibilities, as well as the possibility of being subject to legal limitations, limits, and fines. This is required for national security, territorial disturbance, or crime, moral protection, the preservation of one's reputation and the rights of others, the prevention of exposure of secret information obtained, and the maintenance of the judiciary's authority and impartiality.

While the Philippine Constitution protects people's rights to free speech and expression, this

does not give them free license to write to the detriment of others. This means that the right to freedom of expression is not absolute and has limitations.

Defamation

Defamation is the act of providing a false statement about another individual/groups of individuals which unjustly harms their reputation.

Elements of Defamation

1. There must be an imputation of a crime, or of a vice or defect, real or imaginary, or any act, omission, condition, status, or circumstance.
2. The imputation must be made publicly.
3. It must be malicious.
4. The imputation must be directed at a natural or juridical person, or one who is dead. Here the identity of the offended party must be clearly identified.
5. The imputation must tend to cause the dishonor, discredit, or contempt of the person defamed.
6. It must be in writing (for cyber libel; writing must be done through a computer system or any other similar means which may be devised in the future).

Types of Defamation

1. Libel – defamatory statement made in writing, printed words, or images. Libel is “a public and malicious imputation of a crime, or of a vice or defect, real or imaginary, or any act, omission, condition, status, or circumstance tending to cause the dishonor, discredit, or contempt of a natural or juridical person, or to blacken the memory of one who is dead,” based on Article 353 of the Philippine Revised Penal Code.

Note: How do we differentiate libel and cyber-libel?

As to the manner of commission, **ordinary libel** is committed by means of writing, printing, lithography, engraving, radio, phonograph, painting, theatrical exhibition, cinematographic exhibition, or any similar means; **cyber libel** is committed through a computer system, or other similar digital means.

As to the **venue of filing the complaint**, ordinary libel is filed with the RTC of the province or city where the libelous article is printed and first printed or at the place where one of the offended parties reside at the time of the commission of the offense (Art. 360, RPC). As regards cyber libel, it is filed only with the RTC at the place where one of the offended parties reside at the time of the commission of the offense.

As to the **penalty imposed**, ordinary libel shall be punished by *prison correccional* in its minimum and medium periods or a fine ranging from 200 to 6, 000 pesos, or both, in addition to the civil action which may be brought by the offended party. A cyber shall be punished by one degree higher.

2. Slander – defamatory statement that is spoken, oral defamation. In *De Leon v. People*, slander is “libel committed by oral (spoken) means, instead of in writing. It is defined as “the speaking of base and defamatory words which tend to prejudice another in his

reputation, office, trade, business or means of livelihood.'"

Obscene Speech

Obscenity refers to any act that strongly offends the public sense of decency. Obscene speech and conduct portray sexual conduct in an offensive manner; it also lacks literary, artistic, political, or scientific value.

Interpretations of obscene speech are culturally and politically relative.

Hate Speech

There are some statements and views that are illegal to express because of their harmful, offensive, and obscene content that may harm others. Inflammatory speech, or speech that advocates violence and harm unto others, may incite lawless actions against an individual or a group of people.

LEARNING CONTENT (Key Issues on Freedom of Expression)

The Internet allows the worldwide exchange of news, ideas, opinions, rumors, and information. Its wide accessibility, open discussions, and anonymity make it an effective communication tool. People must frequently make ethical considerations about how to exercise such extraordinary freedom and power.

Controlling Access to Information on the Internet

Internet safety or **e-safety** promotes the physical and psychological well-being of internet users, most especially children and adolescents on the internet. Young people are more susceptible to online risk. Applications and internet features for e-safety are made available, such as parental control softwares that limit and monitor children's internet activity.

Anonymity on the Internet

Anonymous expression refers to the expression of opinions who do not identify their name. The freedom to express oneself without fear of repercussions is a fundamental principle in a democratic society. Anonymity is much more vital in countries where free expression is prohibited. However, in the wrong hands, anonymous communication may be used to engage in unlawful or immoral behavior.

Through the freedom and anonymity that the internet offers, one may use it to hurt others by spreading hateful speech. **Cyberbullying** denotes cruel and aggressive acts carried towards an individual or a group of people through the internet. The use of social media has severely highlighted the effects of cyberbullying.

Hate Speech

Persistent or deliberate harassment directed at a specific person constitutes hate speech, which is punishable by law, but general, broad statements expressing hatred of an ethnic, racial, or religious group cannot. A threatening private communication delivered to a person over the Internet, a public remark placed on a Web site stating intent to conduct acts of hatred motivated violence against specific persons, and libel directed at a specific person are all crimes that can be punished.

Pornography

The Internet has aided the pornography business by offering quick, affordable, and simple access to a vast assortment of pornographic Web sites. Consumers of pornography can avoid upsetting others or being humiliated by others watching their purchases thanks to Internet access.

Establishing a computer usage policy that restricts access to pornographic sites, identifying individuals who breach the policy, and acting against those users—regardless of how humiliating it is for the users or how bad it may be for the company—are all reasonable procedures.

POST-TEST 7

Answer the following questions:

1. Identify ways in which speech can be regulated or limited in the classroom. Does this limit your freedom of expression?
2. Fake news is rampant on social media. How do we differ misinformation and disinformation? How does fake news affect one's freedom of expression?

SUMMARY

- Freedom of expression is defined as the right to express one's opinions, and to share and receive information and ideas without interference from public authority.
- The right to freedom of expression is not absolute; it has its limitations, such as defamation, hate speech, and obscene speech.
- Defamation is the act of providing false statement about an individual. **Libel** is a defamatory statement made in writing; **slander** is oral defamation.
- Internet safety is necessary to protect individuals, most especially children and the youth, from online risks.

REFERENCES

- Bychawska-Siniarska (2017). Protecting the Right to Freedom of Expression Under the European Convention on Human Rights – A handbook for legal practitioners. Council of Europe.
- European Convention on Human Rights. (n.d.). Freedom of Expression. <https://www.echr.coe.int/Pages/home.aspx?p=basictexts&c>
- The Constitution of the Republic of the Philippines. (n.d.). Official Gazette of the Republic of the Philippines. <https://www.officialgazette.gov.ph/constitutions/1987-constitution/>
- United Nations. (n.d.). Universal Declaration of Human Rights. <https://www.un.org/en/about-us/universal-declaration-of-human-rights>
- UNESCO (2009). Freedom of Expression, Access to Information, and Empowerment of People. World Press Freedom Day 2008.

Chapter 8 – Social Networking

MODULE OVERVIEW

In this chapter, we are going to look at the concept of social networking and identify the different uses of social networking sites. We will also study the ethical issues on using social networking sites and on the online virtual worlds and its relationship to the individual's societal behavior.

MODULE LEARNING OBJECTIVES

At the end of this, students are expected to:

- Understand the concept of social networking
- Identify the different uses of social networking sites
- Explain the different ethical issues on using Social Networking Sites
- Articulate the ethical issues on immersing to the online virtual worlds based on ethical theories

PRE-TEST 8

Answer the following questions to gauge your knowledge on the topic.

1. What is the difference between social media and online virtual worlds?
2. Do online communities provide a safe space for anonymity? Why or why not?
3. How is societal behavior affected by social networking?

(LEARNING CONTENTS) Social Networking

Social networking sites establish an online community of Internet users which allow members to break down barriers due to time, distance, and cultural differences; such a site enables social interaction with others online. It enables people to communicate with one another online by sharing their thoughts, insights, information, hobbies, and experiences. Members of an online social network may communicate with friends, family members, and colleagues—people they already know—but they may also utilize the site to build new personal and professional relationships.

With over 2 billion Internet users globally, there are an infinite number of interests represented online, as well as a proportionally enormous number of social networking Web sites catering to those interests. There are thousands of social networking Web sites.

The following are some of the more popular social networking websites.

Social Networking Website	Description	Estimated unique visitors monthly
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 Facebook*	A social networking platform where you may remain in touch with friends, publish images, share links and videos, and meet new people online.	750 million
 Tumblr	Microblogging platform and social networking site that allows users to post shortform blogs.	300 million
 Twitter	Real-time information service for friends, family members, and employees who want to stay in touch by exchanging 280-character messages.	250 million
 LinkedIn	Business-oriented social networking site used for professional networking.	110 million
 	Members of this social networking site can pin photos, movies, and other stuff to their pin board to share with others.	86 million
 Instagram*	Online photo- and video-sharing social networking service.	59 million

*Facebook and Instagram are currently under the same company, Meta.

LEARNING CONTENTS (Uses of Social Networking Sites)

Although social networking sites are generally utilized for non-business purposes, a number of forward-thinking businesses are using this technology to promote, evaluate job applications, and offer products and services.

A growing variety of business-oriented social networking sites are being developed to

encourage and support relationships with consumers, clients, potential workers, suppliers, and business partners worldwide.

Social Network Advertising

Social network advertising involves the use of social networks to promote products and services. Advertisers were quick to grasp social networking's potential as another channel for promoting products and services. It is believed that 89 percent of advertising promote their products using free tools such as Facebook, Twitter, or Pinterest. Furthermore, 75% of advertisers employ paid media, such as Facebook advertisements or sponsored blog material.

Social network advertising has two significant advantages over more traditional advertising media (e.g., radio, TV, and newspapers): (1) advertisers can create an opportunity to generate a conversation with ad viewers, and (2) ads can be targeted to reach people with the desired demographic characteristics.

There are different social network advertising strategies that companies may utilize:

- **Direct Advertising.** Placing banner ads on a social networking site is an example of direct advertising. An ad can be shown to every visitor to the website, or it can be aimed toward those members who are most likely to be interested in the product based on information in user profiles.
- **Advertising Using an Individual's Network of Friends.** Companies can use social networking sites to advertise to any individual's network of contacts.
- **Indirect Advertising Through Groups.** Innovative businesses are also using a marketing technique by forming groups on social networking Web sites that interested users can join by becoming "fans," and these groups can quickly grow in terms of fan count to become a marketing tool for a company looking to market contests, promote new products, or simply increase brand awareness.
- **Company-Owned Social Networking Web Site.** Companies can form their own social networking sites. As an example, Dell created IdeaStorm, a social networking sites for their customers and users, in February 2007. This is used for its millions of customers to talk about the company's new products, and the developments they want to see.
- **Viral Marketing.** Viral marketing encourages individuals to spread a marketing message to others, potentially increasing the message's visibility and effect exponentially when one person informs two people, each of those two people tells two or three more people, and so on. A viral marketing campaign's purpose is to create a buzz about a product or idea that spreads quickly and widely.

The Use of Social Networks in the Hiring Process

According to a 2012 survey on human resource and company hiring, 92% of the respondents use or plan to use social media such as Facebook, LinkedIn, or Twitter in their recruitment process. Employers can and do look into job prospects' social networking sites when making recruiting decisions. Information from these sites include one's gender, age, marital status, sexual orientation, religion, and political affiliation.

The Use of Social Media to Improve Customer Service

Market research has now improved through the use of social networking sites. Consumers share their experiences online, such as their opinions on assorted products and services. Consumers also use social media to ask for advice on better usage of products/services. Companies can monitor these to and respond to consumers; through this, customer service are done online. Customers and companies can resolve issues and questions about certain products/services through social networking sites.

Social Shopping Web Sites

Social shopping sites combine two of the most popular internet activities: shopping and social networking. Members of social shopping Web sites can often create their own pages to collect information and photographs about things of interest. The majority of social shopping websites make money through retailer advertising. Some also make money by providing shops with information about their members' likes and dislikes.

LEARNING CONTENTS (Social Networking Ethical Issues)

When you have a community of tens of millions of members, not everyone will be a nice "neighbor" and follow the community rules. Many people will push the boundaries of socially acceptable behavior. Cyberbullying, cyberstalking, interactions with sexual predators, and the uploading of inappropriate information are all typical ethical difficulties for members of social networking Web sites.

Cyberbullying

- Cyberbullying is defined as the harassment, torment, humiliation, or threatening of one minor by another minor or group of minors using the internet.
- Cyberbullying can be difficult to identify and stop since it can take many forms. Minors should notify their parents if they become a victim of cyberbullying, but this does not happen very often.
- When school officials intervene in an attempt to reprimand pupils for cyberbullying, they are often sued for infringing on the student's right to free expression, especially if the behavior happened outside of school grounds. As a result, several schools have changed their disciplinary policies to reserve the power to punish a student for actions committed off school grounds if they jeopardize a kid's safety and well-being while in school.

Cyberstalking

- Cyberstalking is defined as threatening behavior or unwanted advances directed at an adult via the Internet or other kinds of online and electronic communication; it is the adult equivalent of cyberbullying.
- Cyberstalking is not uncommon to escalate into abusive or excessive phone calls, threatening or obscene letters, trespassing, vandalism, actual stalking, and even physical assault.

Encounters with Sexual Predators

- Some social networking Web sites have been criticized for not doing enough to protect minors from encounters with sexual predators.
- MySpace spent two years purging potential problem members from its site, including 90,000 registered sex offenders banned from the site in early 2009.

Uploading of Inappropriate Material

- Most social networking platforms prohibit the sharing of videos portraying violence or obscenity. The majority of other social networking websites have terms of service agreements, privacy policies, or a content code of conduct that describe essential legal aspects of the website's use.
- In most cases, the terms state that the website reserves the right to remove content and terminate user accounts that violate the site's policies. The policies set specific limits on content that is sexually explicit, defamatory, hateful, or violent, or that promotes illegal activity.

LEARNING CONTENTS (Online Virtual Worlds and its relationship to the individual's societal behavior)

An **online virtual world** is a shared multimedia, computer-generated environment in which users, represented by avatars, can act, communicate, create, retain ownership of what they create, and exchange assets, including currency, with each other.

1. Massively Online Role-Playing Games (MORPG)

- One type of online virtual world, a massively multiplayer online game (MMOG), is a multiplayer video game capable of supporting hundreds and even thousands of concurrent players.
- The games are accessible via the Internet, with players using personal computers, game consoles, and even smartphones. Massive multiplayer online role-playing games (MMORPG) is a subcategory of MMOG that provides a huge online world in which players take on the role of a character and control that character's action. Characters can interact with one another to compete in online games and challenges that unfold according to the online world's rules and storyline.
- Avatars use the virtual world's currency to purchase goods and services in the virtual world. The value of objects in a virtual world is usually related to their usefulness and the difficulty of obtaining them.
- Avatars can also earn virtual world money by performing tasks in the virtual world, or their owners can purchase virtual world money for them using real world cash.

2. Multiplayer Online Battle Arena (MOBA)

- This is a subgenre of strategy video games where two teams of players compete on a predefined battlefield. Each player controls a single character with a set of unique abilities that level up during the course of a game and contributes to the team's overall strategy.
- MOBA is a genre of electronic games that mixes elements of action, strategy and RPG (Role *Playing* Game). The acronym MOBA comes from the English "Multiplayer *Online Battle Arena*" in which players divide into teams and play matches that last on average between 30 minutes to 1 hour.
- Despite having some similarities, the MOBA genre should not be confused with *massive multiplayer online* (MMO), in which thousands of players interact simultaneously in an open online world, controlling a single character during an indeterminate campaign.
- Thus, while the MMO is marked by "no end", the MOBA has a maximum number of players and is defined in short matches in which a team win. Once matches are completed, players do not transfer items or characters to the next ones.
- MOBA games have become extremely popular all over the world and are already trading millions of dollars a year in the *eSports* industry. MOBA game competitions have the highest awards in the industry and have even distributed nearly \$ 25 million in a single Dota 2 championship in 2017.

Crime in Virtual Worlds

The freedom and anonymity provided to users (and subsequently, their avatars) in a virtual environment enables some people to unleash their darker side. Thus, virtual worlds offer a number of intriguing problems about what constitutes a criminal conduct and whether law enforcement—real or virtual—should become engaged in activities that occur in virtual worlds.

Some virtual acts are clear violations of real-world laws and must be reported to law enforcement authorities, such as avatars dealing in drug trafficking or stolen credit cards. Other virtual behaviors, such as online muggings and sex crimes, might create real-life suffering for the human owners of the avatars involved, but do not typically rise to the level of a real-life crime. Regulations in virtual worlds are dependent on game administrators; internal consequences include getting kicked out of games. However, these regulations are minimal in comparison to how they should be dealt offline.

Educational and Business Uses of Virtual Worlds

Virtual online worlds are also being used for education and business purposes. The New Media Consortium (NMC) is an international consortium of hundreds of colleges, universities, museums, and research centers exploring the use of new media and technologies to improve teaching, learning, and creative expression. Members of NMC can conduct classes and meetings from within a growing number of virtual learning worlds.

POST-TEST 8

Exercises:

- f. Answer the following
4. What is a social networking site?
 5. Enumerate two social networking sites that you are using regularly and how it is beneficial to you.
 6. Select one social networking ethical issue that you think happens more often? Explain why.
 7. What is an online virtual world?
 8. What is the difference between MORPG and MOBA?
- g. Answer the following
- a. A(n)_____brings shoppers and sellers together in a social networking environment in which participants can share information and make recommendations while shopping online.
 - b. _____encourages individuals to pass along a marketing message to others, thus creating the potential for exponential growth in the message's exposure and influence as one person tells two people, each of those two people tell two or three more people, and so on.
 - c. The two primary objectives of social media advertisers are driving traffic to a Web site to increase product sales and raising_____.
 - d. _____is the harassment, torment, humiliation, or threatening of one minor by another minor or group of minors via the Internet or cell phone.
 - e. How many people are estimated to be Internet users worldwide?

SUMMARY

- Social networking sites establish an online community of Internet users which allow members to break down barriers due to time, distance, and cultural differences; such a site enables social interaction with others online.
- Business-related social networking sites are used to promote and support consumer relationships, as well as relationships with clients, potential employees, suppliers, and business partners. This also promotes e-commerce.
- Online shopping sites also allows retail businesses to share information and make recommendations to shoppers and sellers.
- Direct advertising, advertising through an individual's network of friends, indirect advertising through social networking groups, advertising via company-owned social networking Web sites, and viral marketing are all examples of social network advertising tactics.
- Cyberbullying is defined as the harassment, agony, humiliation, or threatening of one juvenile by another minor or group of minors using the Internet or cell phone.
- Cyberstalking is defined as threatening behavior or unwanted advances directed at an adult via the Internet or other kinds of online and electronic communication; it is the adult equivalent of cyberbullying.
- An online virtual world is a shared multimedia, computer-generated environment in

which users, represented by avatars, can act, interact, create, retain ownership of what they create, and exchange assets, including currency, is known as an online virtual world.

- Virtual worlds raise many interesting questions about what constitutes a criminal conduct and whether law enforcement, real or virtual, should intervene in crimes committed in virtual worlds.
- Virtual online worlds are increasingly being used for education and business purposes.

REFERENCES

BOOKS:

- Matthew N. O. Sadiku | Adedamola A. Omotoso | Sarhan M. Musa "Social Networking" Published in International Journal of Trend in Scientific Research and Development
- Khan, Ameer. (2015). Ethical Issues in Social Networking. 10.13140/RG.2.1.4289.6080.
- Reynold G.W., (2015), Ethics in Information Technology 5th Edition, Cengage Learning, USA

E-SOURCES:

- (PDF) *Ethical issues in social networking*. (2015, October 10). ResearchGate. https://www.researchgate.net/publication/282701876_Ethical_Issues_in_Social_Networking
- Jamie. (2016, June 16). *101 social networking sites you need to know about in 2021*. Make A Website Hub. <https://makeawebsitehub.com/social-media-sites/>
- Reardon, G. (2019, April 7). *Virtual worlds: Are they good or bad for children?* SWHELPER. <https://swhelper.org/2015/08/31/virtual-worlds-are-you-an-optimist-or-a-pessimist/>
- *Social networking sites*. (n.d.). ScienceDirect.com | Science, health and medical journals, full text articles and books. <https://www.sciencedirect.com/topics/social-sciences/social-networking-sites>
- *What is a virtual world? - Definition from Techopedia*. (2011, October 11). Techopedia.com. <https://www.techopedia.com/definition/25604/virtual-world>
- *What is MMORPG (Massively multiplayer online role playing game)?* (2019, August 2). Computer Hope's Free Computer Help. <https://www.computerhope.com/jargon/m/mmorpg.htm>
- *What is multiplayer online battle arena (MOBA)*. (n.d.). IGI Global: International Academic Publisher. <https://www.igi-global.com/dictionary/multiplayer-online-battle-arena-moba/99945>

Chapter 9 –The Impact of IT on Productivity and Quality of Life

MODULE OVERVIEW

In this chapter, we will discuss the influence of information technology on productivity in several spheres of life. We will also examine our country's progress in bridging the digital gap.

MODULE LEARNING OBJECTIVES

At the end of this, students are expected to:

- Understand the impact of Information technology to the standard of living and worker's productivity
- Determine the status of the country in addressing Digital Divide
- Understand and explain the impact of IT in the different aspect of human life

PRE-TEST 9

Answer the following questions to gauge your knowledge on the topic.

1. The spread of information technology has provided digital productivity in every aspect of life. Provide some example of how IT is used to increase productivity.
2. What are some examples of digital divide in the country?

LEARNING CONTENTS (The Impact of IT on the Standard of Living and Worker Productivity)

The level of life varies widely amongst groups within a country as well as between nations. Gross domestic product (GDP) per capita is the most commonly used indicator of a person's material level of life. National GDP is the total annual output of a country's economy. Overall, developed countries have a higher standard of living than developing countries.

IT Investment and Productivity

Productivity is defined as the amount of output generated per unit of input and may be quantified in a variety of ways. For example, at a factory, productivity may be measured by the number of work hours required to make one item, but in a service sector firm, productivity may be measured by the yearly revenue generated by an employee divided by the individual's annual wage. Most countries have been able to create more products and services throughout time, not by increasing input proportionally, but by improving production efficiency. Because the average hour of labor generated more products and services, productivity advances have resulted in rises in the GDP-based standard of living.

IT has played an essential role in allowing innovation, which is a crucial driver in increasing

productivity. IT, as well as other new technology and capital expenditure, is used by progressive management teams to execute improvements in goods, processes, and services.

Productivity gains were simple to quantify in the early days of information technology, in the 1960s. Midsized businesses, for example, sometimes employed a dozen or more accountants devoted completely to payroll accounting. When firms deployed automated payroll systems, they required fewer accounting personnel. The benefits of such IT expenditures in terms of

productivity were evident.

Organizations are currently attempting to upgrade IT systems and business processes that have already undergone numerous rounds of improvement. Organizations are also introducing new IT capabilities to assist workers who already have a variety of personal productivity programs on their desktop computers, laptops, and cellphones. Instead of removing people, IT advancements save workers tiny amounts of time each day.

The Digital Divide

The digital divide refers to the gap between people who have and do not have access to contemporary information and communications technologies, such as cell phones, smartphones, personal computers, and the Internet. There are around 2.4 billion Internet users globally, however the distribution of Internet users varies substantially by area.

The digital divide extends between nations and even within countries, separating individuals of different ages, socioeconomic backgrounds, and those who reside in cities vs those who live in rural regions.

High-low Internet penetration by country within region

Region	Country with highest Internet penetration	% of population	Country with lowest Internet penetration	% of population
Africa	Morocco	51%	Ethiopia	1%
Americas	Falkland Islands	96%	Belize	23%
Asia	South Korea	82%	Myanmar	1%
Europe	Monaco	100%	Kosovo	20%
Middle East	Qatar	86%	Iraq	7%
Oceania/Australia	Australia	89%	Papua New Guinea	2%

Source Line: Internet World Stats, www.internetworldstats.com/stats.htm.

For a variety of reasons, many individuals think that the digital divide must be addressed. Clearly, if a person in distress had simple access to a communications network, health, crime, and other issues might be dealt more swiftly. Access to information and communication technology may also improve learning and give a plethora of educational and economic opportunities, as well as alter cultural, social, and political conditions. The Internet is

progressively providing individuals with the information they need to manage their careers, retirement, health, and safety.

How Philippines catch up?

The *Philippines Digital Economy Report 2020* of the National Economic Development Authority and the World Bank highlighted that the Philippines still experiences “a very significant digital divide”, with more than half of the homes in the country without Internet connectivity and fixed and mobile Internet penetration in the Philippines being relatively low in comparison to its Southeast Asian neighbors. This disparity is thought to contribute to uneven access to Internet-delivered services.

The digital divide in the Philippines might be linked to a lack of telecommunications infrastructure. It is an issue that has persisted owing to a combination of factors, one of which is the government's inability to invest in digital infrastructure. Bureaucracy involved in obtaining permissions from local governments by private enterprises have also added to this issue.

As a solution, the Department of Information and Communications Technology is spearheading the development of a National Broadband Network, the first draft of which revealed that the government will prioritize the development of Internet access in rural areas, particularly in areas where web access is scarce or very limited.

Under the first draft of the National Broadband Plan, key cities and provinces—particularly the rural areas where commercial telecommunications services are absent—will have Internet access by 2022.

These areas are based on the National Economic and Development Authority's (Neda) National Spatial Strategy for 2017 to 2022 and the National Telecommunications Commission's (NTC) Fixed and Wireless Broadband Data Survey.

Government Initiatives and Programs to enhance the ICT Literacy of the Citizens

The government recently directed the Department of Information and Communications Technology (DICT)'s Literacy and Competency Development Bureau (ILCDB), Cybersecurity Bureau, and Regional Offices to develop and implement webinars aimed at capacitating and upskilling the ICT competencies of teachers, administrators, educators, and students in various public and private educational institutions, as well as state universities and colleges (SUCs).

Major programs and projects of DICT that aims to improve ICT literacy.

e-Filipino

As part of the United Nations' Millenium Development Goal of delivering ICT access to everyone, the DOST-ICTO uses ICT to foster community growth and development in the Philippines through building Community eCenters (CeCs). These centers cater to the ICT demands of many individuals by providing alternative learning and skill training

zo each community. It also intends to equip the special needs sector for impact sourcing and ITpreneurship by providing job opportunities.

National Connectivity

Globally, connectivity is acknowledged as a growth driver. According to a World Bank research, every 10% increase in broadband penetration resulted in a 1.27 percent boost in a developing country's GDP (GDP). However, many sections of the nation and many Filipinos remain disconnected in some form. This issue is addressed in the software.

e-Government Harmonization

e-Government (eGov) is critical to the success of the MDP and the President's Social Contract. The Philippines is now ranked 95th in the world in terms of eGovernment. This initiative intends to raise our ranking to 50 or higher by tackling critical challenges such as interoperability and efficiency through legislative innovations as well as strategic infrastructure improvements.

e-Civil Servants

The ICT Office is responsible for administering and managing ICT Competency-based Training Programs in partnership with public and private training institutions, as well as Subject Matter Experts (SME's). The courses are divided into three categories: e-Government Management, Application Development, and Technology Solutions. This initiative also intends to improve the digital capabilities of government personnel.

ICT Enabled Startup

This is a new program that encourages the spirit of entrepreneurship and leverages the use of ICT to effect change and give answers to challenges in many sectors and society at large by assisting in the formation of technology companies.

The ICT Office aspires to be the government's leader in strengthening the local technology startup community toward a common goal: to be globally recognized as an active startup hub in Asia and the world, and to effectively mobilize its key resources and players to create and nurture both domestic and foreign startups.

LEARNING CONTENTS (The Impact of IT on Different Aspects of Human Life)

Healthcare Information Technology (HIT)

The healthcare industry has seen a considerable transition as a result of information technology. Information technology is rapidly revolutionizing the healthcare business, from electronically checking in patients and updating their medical records to digitally transmitting blood test results to patients.

While information technology refers to the use of computers, telecommunications, and other systems to store, retrieve, and distribute data, healthcare information technology refers to the secure use of technology to handle health-related data.

E-prescriptions, electronic health records (EHRs), and other tech tools that assist people reach health objectives such as monitoring blood sugar levels or quitting smoking are the most prevalent examples of healthcare information technology. Information technology has opened the road for more accurate EHRs/EMRs, allowing patients to get rapid and simple access to a variety of healthcare facilities. Furthermore, through numerous mobile apps and information platforms, it has given patients more control over their health.

Healthcare information technology is significant because it:

- Helps in delivering more accurate, actionable, and accessible information related to a patient's health that can be customized to meet the individual's needs.
- Allows better and faster decisions related to health risks that affect an individual as well as the public.
- Supports communication between patients and healthcare professionals and helps in decision-making.
- Helps build networks of social support for both patients as well as healthcare professionals.
- Improves awareness among patients as well as the general public about health-related matters that can lead to positive outcomes.

Impacts of ICT on Mobility

The potential implications of ICT on mobility include a change (changing origin, destination, method, or route), a substitution (from a physical flow to an information flow), or the creation of a movement. Such a result is dependent on the social and geographical environment in which migration occurs. The following summarizes the projected effects of ICT on the transportation sector.

Transport substitution

The rapid proliferation of ICT, such as mobile phones, intranets, and teleconferencing, promotes new forms of mobility and the possibility to substitute mobility. One of the projected key benefits of ICT is reduced vehicle usage, since it is envisaged that replacement would occur or that vehicle assets will be used more efficiently. However, substitution remains a minor factor in people's physical mobility. ICT has enabled new types of non-physical interactions. The most significant substitution effect has been on postal services, where online communication and tele-consumption have been linked to a significant decrease in physical letter quantities and accompanying transportation operations.

While there were around 268 billion postal pieces sent in the United States in 1990, this amount has now plummeted to 146 billion. Another important form of transport substitution concerns e-commerce where for an online transaction, in-store purchases have been substituted by home deliveries.

Navigation and tracking

While navigation devices have been available for some time, the combination of global positioning systems, wireless communication technologies enabling internet access, and mobile computing devices permitted sophisticated kinds of dynamic navigation and tracking. Navigation aid and real-time traffic information give accurate travel time estimates and the option of alternate routes in the event of disruptive incidents. It is estimated that one billion drivers use navigation applications globally. Overall, this has resulted in significant time and fuel savings for road users, both for passengers and freight transport. Assisted navigation also introduces new issues, such as distributing vehicles on local streets, which can worsen local congestion, and coping with disturbances like school opening hours.

Vehicles and consignments may now be tracked, allowing them to better forecast arrival and delivery dates and save fuel use. It is now feasible to monitor the status of temperature-sensitive commodities throughout transportation. Tracking, as an inventory management approach, also enables more flexible transportation utilization because a consignment may be varied or re-routed if demand changes. Industry standards such as Track and Trace (T&T) are being implemented for containers. It specifies the procedures, data, and interface standards that enable stakeholders like as carriers, shippers, and third-party logistics service providers to communicate with and track containers and their shipments across several modes.

A prospective development concerns self-driving vehicles, that will only be economically feasible if they are given a significant quantity of real-time information about the environment they travel in via on-board sensors as well as information feeds supplemented by ICT. The widespread adoption of self-driving vehicles would have a significant impact on the transportation system, reducing the number of vehicles required to meet current demand, improving mobility for those with physical or financial limitations, lowering the risk of accidents, and improving the time and cost performance of passenger and freight flows alike.

Mobility as service markets

Many transportation markets could previously only be booked through intermediaries such as travel brokers or freight forwarders. With the advancement of the internet, customers may now book transportation services such as air, rail, and bus travel directly, a concept known as mobility as a service. These improvements have been significant for the airline business, increasing competition and the ease of air travel with the ability to check-in online and even carry a virtual boarding pass on a mobile device. A comparable trend has occurred with rail services, notably high-speed rail, but it is also making inroads into the mass transportation industry, including bike rental services. An indirect advantage

is a reduction in the usage of resources and energy for the issuance of travel documents such as passports.

Transport asset management

Ride-sharing and other ICT-enabled transportation services result in improved management of cars, routes, and assets (higher load factor, more trips per vehicle, fewer vehicles for the same capacity, etc.). This is especially true in freight distribution, where

the use of logistics enables improved inventory management and more dependable delivery. ICT has a wide range of applications in transportation asset management. The first is about appointment systems for terminals (ports and rail yards) and distribution hubs. Users can book an access time slot to the facility via an online portal, which should increase the efficiency of both the terminal and the vehicle assets.

Without considerable ICT assistance, including continuing automation of modes, terminals, and distribution facilities, just-in-time inventory management, which minimizes inventory needs, would not be conceivable. Blockchains can help to handle the complexity of intermodal transportation networks by allowing for improved mode and terminal synchronization. Yield management and congestion pricing are also commonly used to better manage available capacity in high-demand situations, but this requires reliable real-time information on demand and supply characteristics. For example, in the case of yield management, an airline might dynamically modify the value of its seats and even ask booked passengers to delay their flight in return for compensation.

Impact of Information Technology on Education

The combination of education and technology has long been seen as the primary driver of human development. Education feeds technology, which in turn feeds education. As a result, it is clear that information technology has influenced changes in the methodology, purpose, and perceived potential of education.

Changes to learning

The ability to access enormous databases of information significantly alters education since learners may now be makers and participants in the access and development of information discourses. Young people can acquire cultural capital from their grasp of new information technologies as a result of their technical literacy, and so have an impact on educational transformation. The same technology also enables the quick interchange of information among researchers on specialized themes, resulting in a significant boost in the pace of information diffusion. Students now have more access to massive volumes of data, which need assistance in selecting, assessing, and analyzing information, as well as learning how to determine the currency, validity, and authenticity of the material itself. All of these shifts in learning have ramifications for classroom instruction.

Changes to teaching

The highest level of change occurring in relation to information technology and education is in the way teaching is increasingly being seen as occurring via the medium of

technology, rather than utilizing technology as an additional extra in the classroom. Information technology particularly impacts course content and teaching methodology and the recruitment and training of teaching staff as well as the content of courses. Information technology requires teachers to learn new sets of skills. Utilizing computer technology improves the educational experience of the students – not so much because of the media itself, but because software programs require teachers to think laterally and systematically, and produce better teaching materials.

Information Technology and the purpose of education

While education in the past has been centered on teaching and learning, information technology has affected changes to the aims of education, therefore now education is increasingly perceived as the process of creating, preserving, integrating, transmitting and applying knowledge. The perceptions of knowledge itself have also changed whereas knowledge could once have been perceived as unchanging, it should now be perceived as “revisionary, creative, personal and pluralistic”. The future of education is not predetermined by modern information technology, but rather that this “future will hinge prominently on how we construct (and construe) the place of technology” in the education process. We are moving from “just-in-case” education to “just-for-you” education” where education is targeted to meet the needs of individual students.

Information Technology and the potential of education

Information technology frees education institutions from the constraints of space and time, and enables the delivery of education services anywhere, anytime. Therefore we can foresee a future where physical libraries would be replaced by digital libraries available to anyone; and that scholars could cease to be located around a geographical focus and will probably become increasingly “located” around a specialization, but physically located anywhere in the world. We could also imagine a day when modern technology will enable students in a given location to access the best of teachers in a given field and to interact with them, whether “live” or via video.

Changing the educational institution

The sheer scope of change underway in communication technology, with changes to the methodology, and modes of education suggests that the educational institution itself may need to be revised at the organizational level as well. Therefore we could foresee a future of increased competition and alliances in which education institutions avoid monolithic approaches to education, and embrace more strategic and collaborative approaches.

Impact of Information Technology on Business

Information technology has dramatically transformed the lives of individuals. It provides businesses the scope to analyze data and plan business strategies accordingly. Utilizing information technology means that the data analysis is accurate, thus optimizing profits.

Information technology has had a major impact on various aspects of businesses. Here are some.

Cloud Computing.

The concept of cloud computing is immensely popular among businesses owing to the efficiency in business operations that it provides. Cloud computing utilizes information technology to capitalize on its ability to provide improved agility and time and resource management for businesses.

Automation of Business Processes.

IT aids in the development of automated business operations. This not only helps to reduce operating costs but also saves time. The time saved may be used to focus on other duties, speeding up corporate operations. Billing, tracking metrics, gathering client data, monitoring certain procedures, and so on may all be readily automated. There is a wide range of automation software available for this purpose.

Remote Work.

The use of information technology allows you to remotely access your company's network. As a result, it provides employees with the ability to do tasks even when they are not physically present at the office.

Protecting Information.

Every organization has a massive database including diverse information about business transactions, client information, and so on. IT provides the necessary resources for storing information in a secure manner. By limiting access to a small number of users, virtual storage solutions may keep information safe. Increased security also ensures that these systems are not hacked and that no data is lost.

Providing Customer Satisfaction.

The finest tools for connecting with consumers and fixing their concerns in real time are provided by information technology. It has made email, social media, and other messaging tools available for this purpose. Help desk software can also enhance the customer service experience.

Management of Resources

A business has diverse resources, including financial, human resource, and so on. IT plays a critical role in seamlessly managing these resources. The integration of Enterprise Resource Planning (ERP), for example, has increased the efficiency of numerous corporate operations. ERP is a type of business management software that allows a company to utilize a collection of connected applications to manage and automate numerous business activities. Such software relies heavily on information technology.

Open-Source Software

Information technology has opened the way for a variety of open-source software that allows free use of certain tools by a variety of organizations. The major advantage of open-source software is its adaptability. This enables changes to the source code. This implies that you may tailor its functions to your own needs. Almost any business software has open-source alternatives that are publicly available on the Internet. Using these

might result in several benefits at a lower cost.

POST-TEST 9

Exercises:

Answer the following

9. What is the digital divide? Where does it exist? Why is it important to bridge the digital divide?
10. What are the major impacts of Information Technology to businesses?
11. What is the role of IT in improving workers' productivity?
12. Give an example of a Navigation software and its features.
13. Give at least three examples of a Healthcare Information Technology.

SUMMARY

- The most widely used measurement of the material standard of living is gross domestic product (GDP) per capita profiles, interact with real-life friends, and meet other people based on shared interests.
- Productivity is defined as the amount of output produced per unit of input.
- Progressive management teams use IT, other new technology, and capital investment to implement innovations in products, processes, and services
- The digital divide is a term used to describe the gulf between those who do and those who don't have access to modern information and technology, such as smartphones, personal computers, and the Internet.
- The *Philippines Digital Economy Report 2020* of the National Economic Development Authority and the World Bank highlighted that the Philippines still experiences "a very significant digital divide", with more than half of total households in the country lacking Internet access and fixed and mobile Internet penetration in the Philippines faring relatively low, compared to its Southeast Asian neighbors.
- As a solution, the Department of Information and Communications Technology spearheads the creation of a National Broadband Network, whose initial draft showed that the government will focus on the development of Internet access in the countryside, particularly in areas where Web access is scarce or very limited.
- Healthcare Information Technology refers to the secure use of technology to manage health-related information.
- The potential impacts of ICT on mobility either involve a modification, a substitution, or a generation of a movement.
- Education feeds technology which in turn forms the basis of education.
- Information Technology provides businesses the scope to analyze data and plan business strategies accordingly.

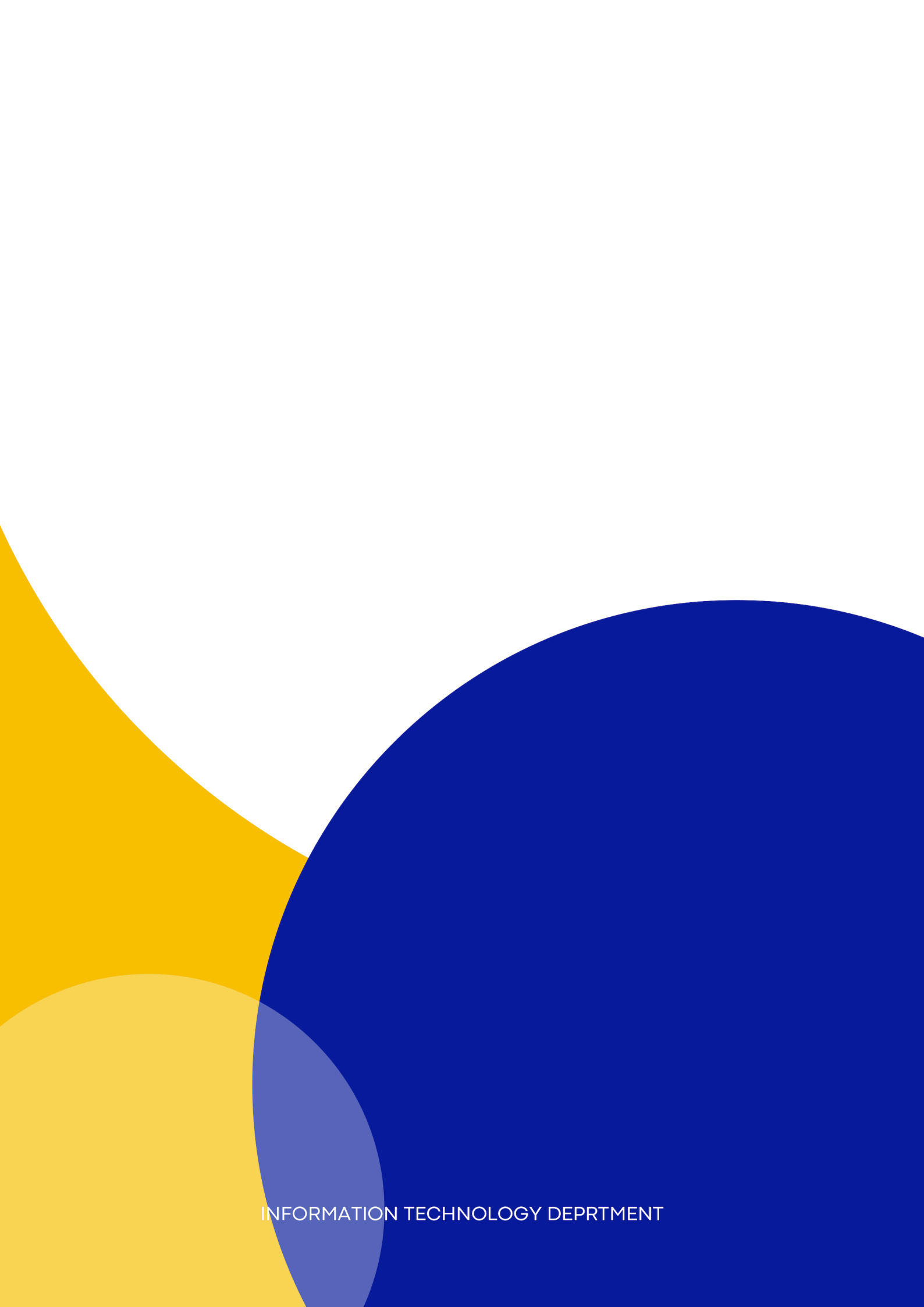
REFERENCES

BOOKS/EBOOKS:

- Reynold G.W., (2015), *Ethics in Information Technology* 5th Edition, Cengage

E-SOURCES:

- “Bridging the digital divide in the Philippines”. 2021. <https://www.bworldonline.com/bridging-the-digital-divide-in-the-philippines/>
- “Bridging the Digital Divide”. 2017. <https://businessmirror.com.ph/2017/04/30/bridging-the-digital-divide/>
- 2.4 – Information technologies and mobility. (n.d.). The Geography of Transport Systems. <https://transportgeography.org/contents/chapter2/information-technologies-and-mobility/>
- The application and impact of information technology in healthcare. (2020, November 20). Fingent Technology. <https://www.fingent.com/blog/the-application-and-impact-of-information-technology-in-healthcare/>
- Programs and projects | DICT. (n.d.). <https://dict.gov.ph/major-programs-and-projects/>

An abstract graphic featuring several overlapping circles. A large blue circle dominates the right side of the frame. On the left, there are yellow circles of varying sizes, some of which overlap with the blue circle. The background is white.

INFORMATION TECHNOLOGY DEPRTMENT